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Advances and Trends in Terrain Classification Methods for Off-Road Perception

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ABSTRACT

Off-road autonomous vehicles (OAVs) are becoming increasingly popular for navigating challenging environments in agriculture, military, and exploration applications. These vehicles face unique challenges, such as unpredictable terrain, dynamic obstacles, and varying environmental conditions. Therefore, it is essential to have an efficient terrain classification system to ensure safe and efficient operation of OAVs. This paper provides an overview of recent advances and emerging trends in off-road terrain classification methods. Through a comprehensive literature review, this study explores the use of sensor modalities and techniques that leverage both appearance and geometry of the terrain for classification tasks. The study discusses learning-based approaches, particularly deep learning, and highlights the integration of multiple sensor modalities through hybrid multimodal techniques. Finally, this study reviews the available off-road datasets and explores the use cases and applications of terrain classification across various autonomous domains. Given the rapid advancements in terrain classification, this paper organizes and surveys to provide a comprehensive overview. By offering a structured review of the current landscape, this paper significantly enhances our understanding of terrain classification in unstructured environments, while also highlighting important areas for future research, particularly in deep-learning-based advancements.

1 | Introduction

Over the past two decades, autonomous ground vehicles (AGVs) have undergone remarkable progress. As researchers continue to push the boundaries of on-road AGV technology, the focus on off-road AGVs has emerged as a critical area of exploration. AGVs have demonstrated a wide array of off-road applications across numerous sectors, including construction sites (Ibanez-Guzman and Malcolm 2002), agricultural automation (Torii 2000; Hassan et al. 2016; Chemhengcharoen et al. 2019; Pour Arab et al. 2024), mining operations (Gutierrez et al. 2004; Androulakis et al. 2020; Zhang et al. 2024), and environmental monitoring (Dunbabin and Marques 2012; Naeem et al. 2008). Military applications, in particular, have embraced the advancements in off-road AGVs for

reconnaissance, surveillance, and logistics operations in complex and unpredictable terrains (Vecherin et al. 2020; Naranjo et al. 2016; Ahmadi et al. 2022).

The industry standard for vehicle autonomy is five levels defined by the Society of Automotive Engineers (SAE). Levels 1 and 2 vehicles include active safety systems that improve driver safety and comfort while the driver is in control. Level 3 vehicles can drive autonomously only in their operational design domain. Vehicles become more autonomous when they reach Levels 4 and 5, using advanced algorithms to negotiate adverse weather conditions and unstructured terrains (Society of Automotive Engineers 2021). The modular anatomy of an autonomous vehicle consists of several key components that work together to enable self-driving capabilities. It starts with

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the perception subsystem, which acquires and interprets data from various sensors, allowing the vehicle to detect and identify obstacles, terrain types, and other elements in its environment. Following perception, the localization and mapping subsystems use this data to create and update maps of the environment, helping the vehicle understand its position within it. The planning component then utilizes these maps to predict the behavior of other objects and plan safe, efficient routes. Finally, the control system translates these high-level plans into actionable commands to navigate the vehicle (Reda et al. 2024).

Perception plays a fundamental role in the operational framework of AGVs as it enables them to detect and understand their surroundings, providing the necessary input for localization, mapping, planning, and control systems to operate effectively. Especially in off-road scenarios, accurately identifying and categorizing the terrain and environment is crucial. The vehicle needs a detailed understanding to adapt its driving strategies to the specific conditions it encounters, thereby improving navigation through difficult and unpredictable landscapes. Dense foliage, changing illumination, rough terrains, and moving obstructions create challenges for precise terrain classification in off-road situations. Off-road AGVs need to navigate through intricate landscapes where traditional road markings and infrastructure are absent, relying heavily on advanced sensor fusion techniques and sophisticated algorithms to overcome the complexities of unstructured environments.

Terrain classification is a crucial aspect of off-road perception, enabling autonomous systems to differentiate between traversable and non-traversable surfaces. While “terrain classification” is often used as a broad term, various analogous concepts—such as terrain awareness (Karnan et al. 2023), scene understanding (Wijayathunga et al. 2023), and traversability estimation (Frey et al. 2024)—frequently appear in the literature. Although these terms emphasize different aspects of environmental perception, they all share the common objective of equipping autonomous vehicles with the ability to interpret their surroundings for safe and efficient navigation. To provide clarity on the scope of this review, terrain classification in this paper refers to the computational methods that process raw sensory data (e.g., images, LiDAR point clouds, radar signals) to determine the navigability of the terrain. These methods serve as the intermediate perception step between sensory input acquisition and motion planning. The output of these approaches can take various forms, including terrain maps, semantic labels, traversability scores, or occupancy grids, all of which contribute to the vehicle's ability to make informed navigation decisions. Due to this diversity in output representations, it becomes challenging to assess all methods using a single unified metric. Instead, the efficacy of a given terrain classification approach is typically evaluated through its corresponding validation results, such as classification accuracy, Intersection over Union (IoU), or F1-score, as well as its real-world performance in enabling successful off-road navigation.

In the context of contemporary terrain classification methods, recent advancements in machine learning, especially deep learning, have been transformative, allowing the development of sophisticated models capable of extracting detailed features from sensor data with high precision. Moreover, the introduction of

semi-supervised and self-supervised learning methods has reduced the reliance on manual labeling of large datasets (Singh et al. 2024; Nie et al. 2022). These techniques use partially labeled or unlabeled data, making the training process more cost-effective and scalable. Additionally, researchers are turning to synthetic data and simulation-based methods to address the scarcity of real-world labeled datasets (Curnis et al. 2022; Nunes et al. 2022; Yan et al. 2022; Le et al. 2021; Małek et al. 2024).

The rapid growth of self-driving and driver-assisted cars has led to significant progress in autonomous vehicle technologies. This development is evident in the increasing number of publications focused on terrain classification and perception, as shown in Figure 1. Alongside these advancements in structured environments, off-road robotics has also seen significant progress, driven by early initiatives, such as the “DARPA Grand Challenge” (DARPA Grand Challenge 2005–2006). This competition, funded by the Defense Advanced Research Projects Agency (DARPA), was one of the first large-scale efforts to push the boundaries of autonomous navigation in unstructured terrain. Another key program, Learning Applied to Ground Vehicles (LAGR) (2004–2008) (Jackel et al. 2006), focused on developing machine learning-based perception systems for off-road environments. These programs led to a surge in innovative terrain classification approaches, many of which set new benchmarks in the field and were documented in academic literature.

Building on this foundation, newer terrain classification methods have since emerged, leveraging advances in deep learning, sensor fusion, and large-scale data set availability. These developments continue to refine off-road perception, further expanding the capabilities of autonomous systems in complex, unstructured terrains. This trend underscores the critical timeliness and relevance of this survey.

There is a concise yet informative body of literature focusing on analogous topics, such as a review of terrain perception methods in autonomous vehicles, warranting acknowledgment of the existing research in this domain. Table 1 offers an overview of various survey papers pertaining to terrain classification methodologies. The survey paper by Papadakis et al. (Papadakis 2013) is the first to survey and categorize terrain traversability methods based on appearance and geometry. However, it is worth noting that terrain classification is treated as a subtopic within the broader context of traversability mapping. As a result, the review

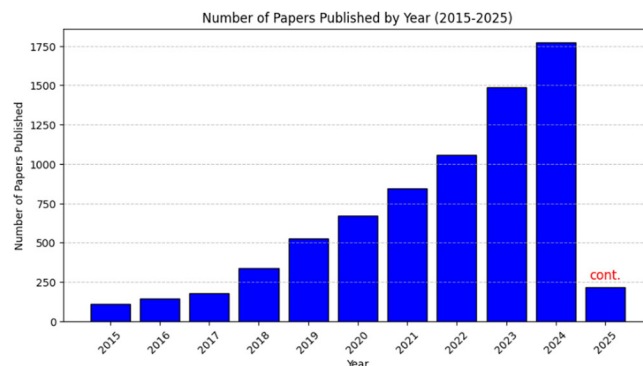


FIGURE 1 | Distribution of relevant publications by year.

TABLE 1 | Summary of existing surveys related to terrain classification.

Authors, year	Number of studies included	Aim of the review	Exploration of TC methods	Comparison of TC methods	Exploration of use cases and applications	Exploration of data set
Papadakis (2013)	92	To offer an organized taxonomy of terrain traversability for AGV motion planning.	◦	×	◦	×
Guastella and Muscato (2020)	102	To provide an overview of recent advancements in learning-based navigation for ground vehicles in unstructured environments.	◦	◦	×	×
Nampoothiri et al. (2021)	106	To review recent advances and challenges in robot design, terrain analysis, and control strategies.	◦	◦	×	×
Beycimien et al. (2023)	171	To survey recent advancements in terrain traversability methods and sensor technologies for ground vehicles.	◦	◦	×	×
Wijayathunga et al. (2023)	166	To review and analyze methods for autonomous robot navigation in unstructured terrains, identifying challenges and future research directions.	◦	×	×	×
Szabó and Weltsch (2024)	14	To provide an overview of challenges in off-road vehicle perception, emphasizing diverse datasets and the role of synthetic data.	×	×	×	✓
Our Survey, 2024	218	To evaluate and categorize terrain classification methodologies, provide a critical analysis, and propose recommendations for future research directions.	✓	✓	✓	✓

Abbreviation: TC = Terrain Classification.
✓: Paper fully covers the topic.
◦: Paper covers the topic partially.
×: Paper does not cover the topic.

did not fully capture the nuances of terrain classification methods, particularly in complex off-road terrains. Additionally, given its publication in 2013, the review does not encompass the latest advancements and state-of-the-art methods in this rapidly evolving field. Beycimen et al. (Beycimen et al. 2023) have contributed to the understanding of ground vehicle terrain traversability for unstructured environments, with a focus on areas such as cost-based traversability, mapping, and partially addressing terrain classification. A recent survey by Szabó et al. (Szabó and Weltsch 2024) overviews the available off-road datasets without addressing the methods of terrain classification. To the best of the authors' knowledge, this survey paper is the first to provide a critical analysis of the latest methods, datasets, use cases, and applications in the field of off-road terrain classification. The authors aim to provide new researchers in the field with a structured and organized overview of the existing challenges in off-road perception and the efforts made to address them.

The key contributions of the review article include:

- To evaluate terrain classification methodologies categorized by appearance, geometry, and hybrid approaches.
- Evaluate the effectiveness and limitations of various terrain classification methods, considering their applicability in off-road autonomous navigation scenarios.
- To offer insights into relevant datasets used and propose recommendations for future research directions in off-road autonomous perception systems.

The rest of the paper is organized as follows: In Section 2, the research methodology is presented, detailing the search and selection strategies. Section 3 presents a taxonomy of terrain classification methods. Section 4 provides a critical analysis of the methods. In Sections 5 and 6, the discussion shifts toward the datasets, applications, and future directions of terrain classification methods.

2 | Review Methodology

Figure 2 represents the methodology used in this survey paper to perform a thorough evaluation and analysis of the existing

research literature. To ensure comprehensive coverage, popular and most relevant publications were selected for use throughout the survey. The process is divided into three sections: the search technique utilized to collect relevant research, the selection criteria employed to screen the studies, and the analysis method used for further review.

2.1 | Search Strategy

The search strategy encompassed a thorough exploration across diverse scholarly databases and search engines, including Google Scholar, Scopus, IEEE Xplore, ScienceDirect, and Deakin University e-library. To refine the search parameters, a keyword co-occurrence analysis was conducted using VOSviewer (Vosviewer—visualizing scientific landscapes 2010) and the Scopus database, facilitating the efficient retrieval and selection of pertinent keywords, aiding in the identification of key terms. Additionally, manual exclusion of certain keywords further streamlined the search process, ensuring a focused and comprehensive review of the literature. Utilizing the identified keywords and Boolean operators, additional searches were conducted across the aforementioned scholarly databases to retrieve the most relevant studies. Additionally, the “connected papers” tool (Connected Papers 2025) was frequently used during the search process to identify and explore additional relevant documents linked to key studies.

2.2 | Selection Criteria

Following the keyword search method, additional screening was required to exclude irrelevant and redundant studies from consideration. The following questions were utilized as criteria to assess every paper:

- Does this paper originate from a peer-reviewed scholarly journal, a conference paper, or a book chapter?
- Is this study written and published in the English language?
- Is this study focused on a method for classifying and analyzing terrain for autonomous vehicles, comparing various

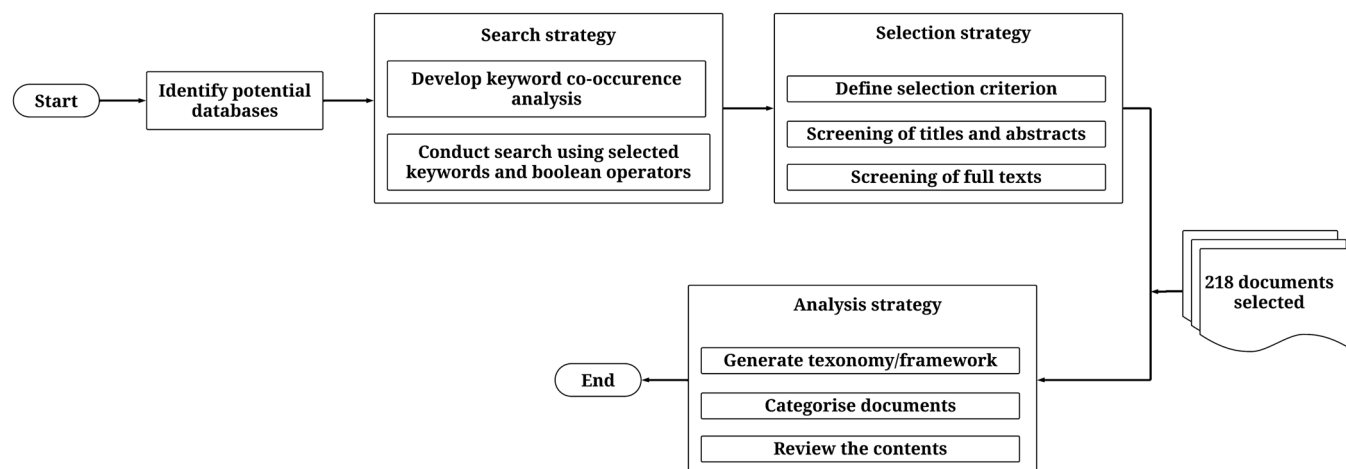


FIGURE 2 | Flowchart of the review methodology used in this study.

methods, presenting a data set, or conducting a survey on the topic?

- Does this paper focus on off-road/unstructured/natural terrain environment?

If all the answers were “Yes,” the paper was included in the survey. Following the search and selection strategies, 218 documents met the final eligibility criteria and were selected for in-depth review and analysis of methods.

3 | Terrain Classification Framework

Traditionally, terrain perception methods were categorized based on the type of sensor used to collect input data (Borges et al. 2022). The sensors can be broadly divided into two categories: proprioceptive sensors, such as IMUs, wheel encoders, motor current sensors, and suspension sensors; and exteroceptive sensors, including cameras, LiDAR, radar, and ultrasonic sensors. Among these, cameras and LiDAR are the most commonly used for terrain classification. While proprioceptive approaches have their advantages, they have been rarely used in isolation in recent years. Therefore, our focus is on exteroceptive and hybrid methods, which we further categorize into a structured taxonomy. Figure 3 presents the categorization of approaches in our proposed taxonomy. We have grouped methods into three main categories:

- **Appearance-based methods**—rely on visual cues and texture information to classify terrain.
- **Geometry-based methods**—focus on the structural and depth information of the terrain.
- **Hybrid approaches**—integration of both appearance and geometry.

These categories are further divided into traditional learning-based and deep learning-based approaches. Additionally, within hybrid approaches, we distinguish between proprioceptive-exteroceptive hybrids and appearance-geometry hybrids, depending on the type of data fusion utilized.

3.1 | Terrain Classification by Appearance

Visual techniques enable the capture of detailed features and textures present in the terrain environment. Recent advancements in computer vision algorithms, machine learning

techniques, and sensor technologies have empowered vision-based methods to attain notable accuracy in analyzing terrain characteristics.

Some autonomous vehicles rely entirely on vision-based perception for scene understanding, foregoing other sensor modalities, such as LiDAR or radar. A notable example is Tesla, which transitioned to a vision-only system for its autopilot and full self-driving (FSD) features in 2021, replacing radar-based perception (Zhang 2024). This shift was driven by advancements in deep learning and camera-based processing, allowing Tesla’s system to infer depth, object motion, and scene context purely from visual data. While such an approach remains a topic of debate, it highlights the growing reliance on vision as a primary modality for autonomous navigation.

3.1.1 | Traditional Learning in Appearance-Based Terrain Classification

In literature, vision-based terrain classification techniques have predominantly utilized binary classifiers to distinguish between road and non-road terrain (Selvathai et al. 2017; Kim et al. 2007). Such approaches are inadequate or overly simplistic for off-road navigation scenarios characterized by diverse terrain types. Commonly employed classifiers in the literature for this task encompass K-nearest neighbors (kNN) (Khan et al. 2011b, 2011a; Yu 2020), support vector machine (SVM) (Chetan et al. 2010b; Omer and Fu 2010; Filitchkin and Byl 2012; Khan et al. 2011b, a), random forest (RF) (Khan et al. 2011b, a; Chetan et al. 2010b; Ono et al. 2015; Zhang et al. 2016; Winkens et al. 2017), decision tree (Khan et al. 2011b, a; Chambers et al. 2015), and Bayesian classification (Khan et al. 2011b, 2011a; Kim et al. 2007, 2009). However, researchers have augmented the accuracy of these baseline classifiers through various methods. These include contrast enhancement techniques proposed by Kim et al. (Kim et al. 2009), utilization of superpixels (Kim et al. 2007; Winkens et al. 2017), integration of gradient features (Omer and Fu 2010), and the development of a partition algorithm by Chetan et al. (2010a), who enhanced baseline classifier accuracy through the incorporation of a temporal label transfer method.

Over the years, researchers in vision-based terrain classification have dedicated significant efforts to determining the optimal types and characteristics of image features to achieve maximum classification accuracy. Descriptors, serving as the backbone of

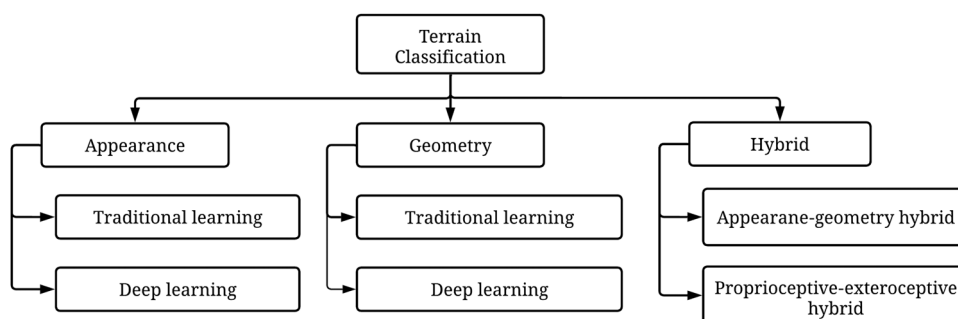


FIGURE 3 | Framework for categorizing terrain classification methods.

image classification, encode crucial information essential for distinguishing between images belonging to different classes. Traditionally, color histograms and texture descriptors have been commonly applied in onboard systems (Dahlkamp et al. 2006; Upcroft et al. 2008). Terrain types in off-road environments can change rapidly, with varying surface textures, lighting conditions, and colors, making color-only and texture-only descriptors insufficient to handle these variations effectively. To overcome these issues, researchers have explored methods to represent texture information using filter banks (Varma and Zisserman 2002), bag of visual words (BOVW) (Filitchkin and Byl 2012), and Markov Random Fields (MRFs) (Vernaza et al. 2008), which show promising precision in classification tasks. However, these methods often come with significant computational overhead, requiring large processing power and memory resources, making them impractical for real-time applications in mobile systems. As a result, efforts have shifted toward more efficient solutions, leading to the proposal of compact composite descriptors (CCDs) (Chatzichristofis and Boutalis 2008), which combine both color and texture in a single global image feature. CCDs offer a more compact representation, reducing the computational burden while still capturing essential information about the terrain. In a comparative analysis by Khan et al. (Khan et al. 2011a), SURF (Bay et al. 2006) and DAISY (Tola et al. 2010) descriptors were compared, with SURF demonstrating superior accuracy levels (99.2%). Previously, Filitchkin et al. (Filitchkin and Byl 2012) achieved remarkable success by classifying terrain using BOVW generated from SURF features for LittleDog, attaining an impressive 100% classification accuracy. Zou et al. (Zou et al. 2014) explored ten color and texture descriptors alongside three classifiers for visual terrain classification. Their experiments underscored the efficacy of composite descriptors, specifically the joint composite descriptor (JDC), showcasing the highest performance among all evaluated combinations. Additionally, JCD proved to be robust to illumination changes, addressing a prevalent issue in color-dependent terrain classification techniques.

In current terrain classification studies, researchers are increasingly combining color and texture-based methods. This strategy takes advantage of color's ability to identify surface properties and texture's capacity to capture surface details. By merging these approaches, classifiers become more resilient to environmental changes and achieve better accuracy in distinguishing between different terrains. In addition to color and texture-based methods, spatial features have also been investigated in some literature (Sung et al. 2010; Lu et al. 2009). Table 2 summarizes key approaches that facilitate traditional learning in appearance-based off-road terrain classification.

3.1.2 | Deep Learning in Appearance-Based Terrain Classification

The introduction of fully convolutional networks (FCNs) (Long et al. 2015) marked a significant breakthrough in deep learning, revolutionizing various computer vision tasks, including off-road terrain classification. Compared to traditional methods, deep learning approaches—particularly convolutional neural networks (CNNs)—have demonstrated superior performance in visually distinguishing terrain in unstructured environments due to their ability to automatically learn hierarchical feature representations from raw data (Holder et al. 2016). In terrain classification, deep learning methods are commonly formulated as tasks of drivable area/free space detection (Wu and Liu 2024; Qiu and Jiang 2025) with semantic segmentation being the more prevalent approach (Viswanath et al. 2021; Sgibnev et al. 2020; Jin et al. 2021). Semantic segmentation is essentially a pixel-wise classification approach where each pixel in an image is assigned a corresponding class label, allowing a model to distinguish between different terrain types. It utilizes deep neural networks to capture spatial and contextual information, making it highly effective for complex off-road perception.

A significant portion of deep learning-based visual terrain classification literature follows the supervised learning

TABLE 2 | Traditional learning methods in appearance-based terrain classification.

Reference	Method	Data	Terrain type
Khan et al. (2011b)	RF/SVM/MLP/LIBLINEAR/J48 Decision Tree/naive bayes/kNN with modified SURF and Daisy descriptors	RGB, grayscale	Grass, asphalt, gravel, big tiles, small tiles
Filitchkin and Byl (2012)*	SVM classifier with BOVW created from SURF descriptor	RGB	Grass, soil, asphalt, gravel, wood chips, mud
Chetan et al. (2010b)	Enhanced kNN/SVM/RF with a partition-based algorithm	RGB	Road, dirt road, grass, gravel, background
Zou et al. (2014)	ELM/SVM with novel composite descriptors	RGB	Big gravel, small gravel, hard soil, pothole, grass, mulch, dirt
Kim et al. (2009)	Bayesian classification with contrast enhancement	RGB	Soil, small Gravel, gravel, asphalt
Omer and Fu (2010)	SVM with RGB features	RGB	Road, snow, track
Lee and Kwak (2011)	MLP with SURF descriptor	RGB	Sky, grass, soil, tree, gravel, outlier

*Tested in real-time.

paradigm, which requires large-scale, pixel-wise labeled datasets to train models effectively. One of the earliest approaches to address the challenge of data scarcity in off-road terrain classification was proposed by Holder et al. (Holder et al. 2016), where transfer learning was employed to leverage knowledge from pre-trained models. The fundamental idea behind transfer learning is that feature representations learned on large-scale datasets can serve as generic image filters, which can then be fine-tuned for more specialized tasks. In this study, SegNet (Badrinarayanan et al. 2017) was utilized as the segmentation model, requiring approximately 10,000 iterations of pretraining to achieve optimal accuracy – which brings up the challenge of real-world adaptability due to computational demands. A similar approach was taken by Sharma et al. (Sharma et al. 2019), where transfer learning was applied using a model pre-trained on DeconvNet (Noh et al. 2015), with a focus on reducing computational complexity by downsampling the DeconvNet architecture to half its original size.

One of the primary drawbacks of CNN-based models in terrain classification remains their high computational complexity, making real-time deployment on resource-constrained embedded systems challenging. To address this issue, subsequent research in off-road terrain classification has focused on developing lightweight architectures while minimizing the compromise on accuracy. Sgibnev et al. (Sgibnev et al. 2020) employed DeepLabV3 (Chen et al. 2017) with a ResNet-34 (He et al. 2016) backbone, integrating Squeeze and Excitation blocks to achieve an optimal balance between inference speed and accuracy. Similarly, Elander et al. (Elander 2021) explored the trade-off between model depth and efficiency, concluding that reducing the number of layers in a ResNet backbone can enhance real-time performance while maintaining sufficient accuracy. Further advancing computational efficiency, Peng et al. (Peng et al. 2024) introduced a network called Lightweight Boundary-Assisted Segmentation Network (LB-ASN), which achieves high accuracy while keeping the model extremely lightweight. Chen et al. (Chen and Lan 2024) proposed modifications to BiSeNetV2 (Yu et al. 2021) by removing

redundant components and introducing a concatenation module (named MICM) to mitigate information loss in deeper network layers.

Following the success of transformer networks (Vaswani et al. 2017) in natural language processing, vision transformers (ViTs) (Dosovitskiy et al. 2020) have emerged as a powerful alternative to CNNs for visual tasks, with off-road terrain classification also experiencing a surge in development due to this advancement. ViTs leverage self-attention mechanisms, allowing the model to focus on different parts of an image at varying levels of granularity. This improves the ability to capture long-range dependencies and spatial relationships, which are crucial for accurate semantic segmentation in complex environments like off-road terrain. An early example of this application in terrain classification was presented by Singh et al. (Singh et al. 2021), where a semi-supervised approach was proposed using the DINO (Caron et al. 2021) transformer, a self-supervised attention mechanism. DINO employs an automatic data selection algorithm based on depth estimation to choose the most informative images, which significantly reduces the need for annotated data while enhancing model performance. The proposed method (called OffRoadTranSeg) and its working process are presented and described in Figure 4. Building on these advancements, Hassan et al. (Hassan et al. 2024b) integrated a ResNet50 backbone with a transformer-based segmentation called MaskDINO (Li et al. 2023) and a detection head to classify terrain, including free space and background. Recent research by Faykus et al. (Faykus et al. 2024) demonstrated the potential of ViTs for off-road perception. Using EfficientViT (Cai et al. 2023), a hardware-efficient variant, they achieved a 94.44% mIoU for traversable terrain on the CaT data set (Sharma et al. 2022), while offering up to four times faster inference on edge devices compared to CNN models like SwiftNet (Wang et al. 2021). Transformer blocks have also been integrated into hybrid network architectures, such as those with a ResNet backbone, as seen in the work of Hassan et al. (Hassan et al. 2024a), or combined with multilayer perceptrons (MLPs), as demonstrated in (Liu et al. 2024b).

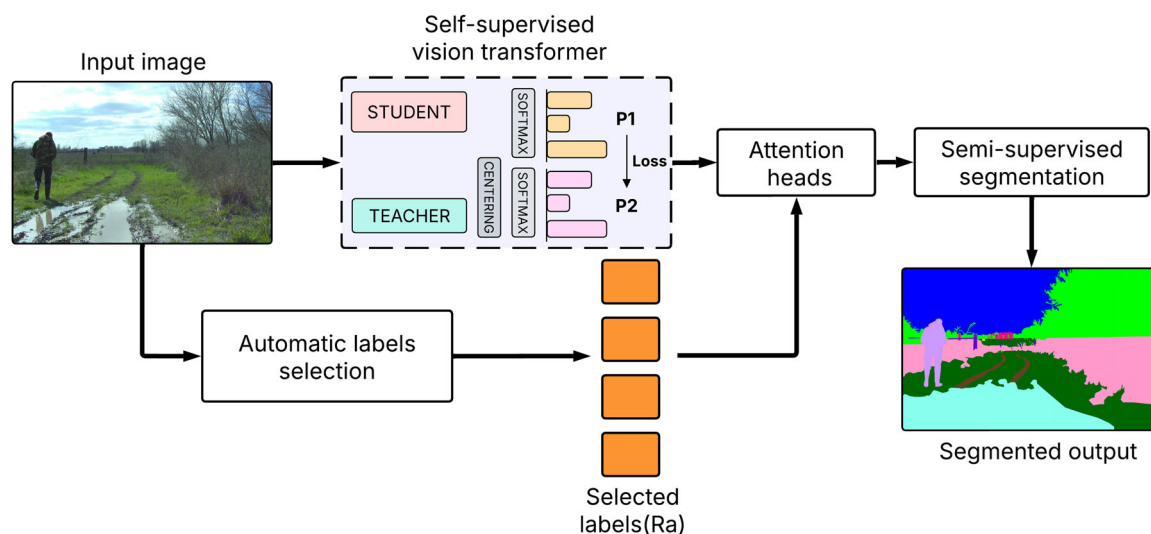


FIGURE 4 | The framework proposed by (Singh et al. 2021) is a two-step approach: Initially, a self-supervised vision transformer is fine-tuned to extract attention maps, followed by an automated selection of labeled images to assess the network's effectiveness in semi-supervised semantic segmentation.

Class imbalance is a significant challenge in both traditional and deep learning-based terrain classification methods, as it can result in biased predictions where the model favors the dominant class (Viswanath et al. 2021). This issue has been addressed in various studies to improve the performance of semantic segmentation in off-road environments. Mayuku et al. (Mayuku et al. 2021) proposed a solution by incorporating multi-resolution local binary pattern (LBP) texture descriptors. As CNNs are sensitive to LBP texture features at different resolutions, fusing RGB and LBP texture features improved segmentation accuracy, particularly in imbalanced datasets. However, tuning texture parameters and the fusion strategy requires extensive experimentation. Shi et al. (2023) assigned class-specific weights to the loss function, thereby improving model performance on less-represented classes.

Off-road terrain classification also faces challenges with illumination variability. Illumination changes can severely affect visual-based methods, leading to poor generalization in real-world conditions. To overcome this issue, Jin et al. (2021) introduced a memory module designed to store significant representations of training images. This module clusters instances of the same class, even under varying illumination conditions, allowing the network to better handle unexpected lighting changes during segmentation tasks. Hanson (2011) explored the use of near-infrared (NIR) images, which are less influenced by lighting variations compared to traditional RGB images. By leveraging both RGB and NIR data, their approach enhances the reliability of semantic segmentation in off-road environments, particularly under challenging lighting conditions. In mobile robotics, the type of vehicle plays a crucial role in determining traversability, as different vehicles may exhibit varying capabilities on off-road terrains. Dabbiru et al. (2021) have annotated and trained datasets based on the vehicle's ability to traverse specific trails, incorporating vehicles like sedans, pickup trucks, and military vehicles for more accurate terrain classification. Guan et al. (2022) proposed GA-Nav that

utilizes coarse-grained semantic segmentation to classify terrain based on navigability levels, leveraging a novel group-wise attention mechanism to focus on features across different terrain types. The operational flow and framework of GA-Nav are presented and described in Figure 5. Another challenge in deep learning-based off-road terrain classification is adversarial attacks—small, deliberate perturbations to the input data that can cause the model to make incorrect predictions. To address this, Deoli et al. (2024b) worked on creating a robustified data set that is resilient to such adversarial attacks, ensuring more reliable model performance in real-world settings. Data limitation is a significant challenge in off-road terrain classification, as datasets are generally smaller compared to large, benchmark datasets like ImageNet or Cityscapes (Cordts et al. 2016). To address this issue, a common approach is to combine multiple datasets, which increases the class pool and improves model performance (Walch et al. 2022; Humblot-Renaux et al. 2022). However, Yang et al. (2022) pointed out that naively merging datasets can introduce label noise, negatively affecting model performance. They showed that uncertainty-aware models outperform traditional segmentation models by quantifying prediction uncertainty. These models generate higher entropy in predictions when tested on new data, improving adaptability. Table 3 summarizes key approaches that facilitate deep learning in appearance-based off-road terrain classification.

3.2 | Geometry-Based Classification Methods

Understanding the shape and geometric properties of terrain is crucial, especially in autonomous navigation. Researchers have developed numerous methods to extract and analyze geometric features from terrain data, enabling the classification of terrain types. In this section, we discuss the trends, principles, and challenges associated with geometry-based approaches to terrain classification.

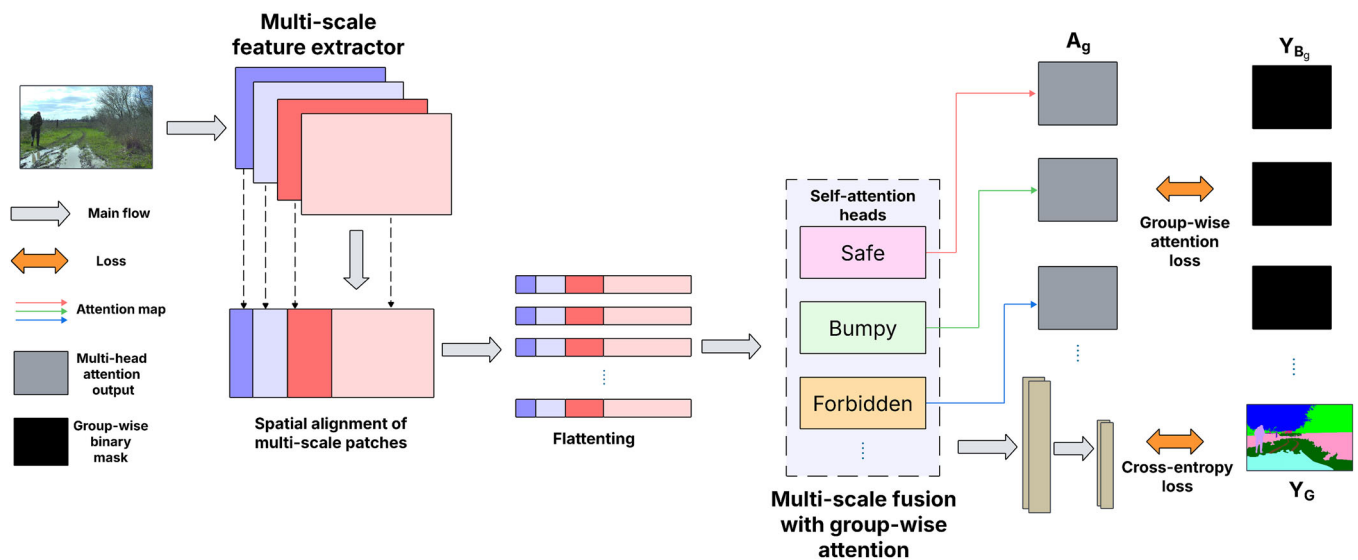


FIGURE 5 | GA-Nav architecture proposed by (Guan et al. 2022): A novel group-wise attention mechanism is introduced for multi-scale feature fusion. Multi-scale features are divided into groups, and the group-wise attention loss enforces alignment between the multi-head attention outputs (A_g) and the corresponding binary ground truth (Y_{B_g}), guiding the network toward more accurate predictions across different feature groups.

TABLE 3 | Deep learning methods in appearance-based terrain classification.

Reference	Method	Data	Terrain type
Wu and Liu (2024)	YOLOP with gated depthwise inception model (DIM)	RGB	Drivable and non-drivable
Shi et al. (2023) ^a	HRNet (Wang et al. 2020) with weighted loss function	RGB	19 terrain classes from RELIS-3D (Jiang et al. 2021)
Jin et al. (2021) ^b	PSPNet (Zhao et al. 2017)/Deeplabv3/DANet (Fu et al. 2019) with MobileNetv2 (Sandler et al. 2018)/ResNet18/HRNet and a Memory Module	RGB	Gravel, dirt, building, vehicle, fence, mulch, tree, pole, grass, bush, rock
Sharma et al. (2019) ^c	Pretrained DeconvNet	RGB	Obstacle, trail/road, sky, grass, tree, vegetation
Sgibnev et al. (2020) ^{a,c}	ResNet34+DeepLabV3 with Squeeze & Excitation Blocks	RGB	Hard ground, soft ground, building, fence, impassable vegetation, passable vegetation, sky, people, vehicle, water, traffic sign, pole, other obstacles, void
Neto et al. (2022) ^{a,c}	Configurable modular semantic segmentation neural network (CMSNet)	RGB	Road, car, person, background
Deoli et al. (2024a) ^b	Semantic segmentation with UNet (Ronneberger et al. 2015) and DeepLabV3	RGB, NIR, Green, Red, REG channel images	Obstacle, sky, trail, grass, vegetation, void
Peng et al. (2024) ^c	Lightweight boundary-assisted segmentation network (LB-ASN) for Multi-task boundary detection and segmentation	RGB	Drivable and non-drivable
Beycimen et al. (2024) ^a	Real-time model switching of 19 Neural Networks	RGB	Sky, vegetation, objects, person, pavement, vehicle, mud, house, asphalt road, smooth road, rough road, very rough road, low traversable grass
Chen and Lan (2024) ^{a,c}	Proposed MICNet by reducing the redundant part of BiSeNetV2	RGB	Drivable and non-drivable
Liyanage et al. (2020) ^d	ResNet18	RGB and HSI	Muddy terrain, grass, trees, water, sky
Faykus et al. (2024) ^d	EfficientViT, SwiftNet, PSPNet w/ResNet-18-34-50-101	RGB	19 Terrain Classes from RELIS-3D

^aTested in real-time.^bHandles illumination effects.^cLightweight architecture proposed.^dAddresses class imbalance.

Geometry-based approaches mostly rely on reconstructing the three-dimensional environment using advanced active and passive sensing technologies such as LiDAR, time-of-flight (ToF) cameras or stereo cameras. These sensors provide rich and precise data about the terrain surface, capturing detailed information about its shape, elevation, slope, and curvature. Among these technologies, LiDAR stands out for its ability to generate high-resolution 3D point clouds (Islam et al. 2022), while stereo cameras offer stereo vision capabilities, enabling the extraction of depth information from pairs of images (Rankin et al. 2009).

Utilizing raw point cloud data for terrain classification significantly enhances the complexity of the process. Hence, the

representation of point clouds plays a crucial role in the outcome. Two prominent approaches include the elevation map-based method (Douillard et al. 2010; Meng et al. 2018; Howard et al. 2006; Shan et al. 2018) and the geometric model fitting method (Douillard et al. 2011; Moosmann et al. 2009; Himmelsbach et al. 2010). The elevation map, often termed as a 2.5D representation, has been a key resource for early researchers due to its simplicity and efficiency. However, Pfaff et al. (2007) highlighted a critical limitation – the inability to represent vertical or overhanging objects in terrain landscapes. To address this, they proposed an extended elevation map by refining data association within the Iterative Closest Point (ICP) algorithm. Subsequently, advancements such as multilevel surface maps and the inclusion of slope information were

introduced to tackle abysses and holes more effectively (Triebel et al. 2006; Joho et al. 2007). Himmelsbach et al. (2010) observed that 2.5D representations often suffer from under-segmentation, especially in complex terrains or scenarios. This issue is linked to dimensionality reduction. To address this, they proposed a method that uses 2D line extraction algorithms to estimate local ground planes from unorganized 3D points. It then compares individual points to the ground plane lines to separate ground from non-ground points reliably. Douillard et al. (2010) proposed a voxel grid-based full 3D segmentation method, where objects are represented by detailed 3D voxels. Although their hybrid elevation map segmentation exhibited superior performance in terms of accuracy, it incurred extended computational time as a limitation. Forkel et al. (Forkel et al. 2021) proposed a real-time probabilistic terrain estimation method by separating terrain and obstacle mapping, ensuring smooth elevation profiles and efficient linear time complexity. Ahtiainen et al. (2017) developed the normal distributions transform traversability mapping (NDT-TM) technique, using 3D LIDAR sensor data to enhance normal distributions transform occupancy (NDT-OM) maps. They further refined the method by training a support-vector machine classifier to distinguish between traversable and non-traversable regions on the NDT-TM maps. Notably, by incorporating intensity distribution and permeability as additional features, they improved classification accuracy, particularly in dense vegetated environments. Broggi et al. (2013) present a purely geometric real-time 3D terrain mapping technique that maps 3D points onto a 2.5D grid, and then iteratively fits them into B-Splines for terrain segmentation. However, the approach may struggle under poor visibility conditions, leading to reduced disparity map density.

Traditional single-planar models for ground segmentation, such as RANSAC (Fischler and Bolles 1987), frequently suffer from under-segmentation, where valid ground points are mistakenly classified as non-ground due to the inability of a single-plane assumption to capture complex terrain variations. Patchwork, introduced by Lim et al. (2021), addresses the under-segmentation issue in LiDAR-based ground segmentation by using ground likelihood estimation (GLE) to classify ground points based on uprightness, elevation, and flatness. Despite its efficiency, Patchwork relies on fixed parameters, making it prone to failures in uneven terrains. To overcome this limitation, Lee et al. (2022) proposed Patchwork++, which dynamically updates segmentation parameters using adaptive ground likelihood estimation (A-GLE). Additionally, it introduces region-wise vertical plane fitting (R-VPF) for multi-layered terrains and reflected noise removal (RNR) to eliminate LiDAR artifacts.

3.2.1 | Traditional Learning in Geometry-Based Terrain Classification

Before the advent of deep learning, traditional supervised learning techniques were fundamental to geometry-based terrain classification (Martínez-Rodríguez et al. 2020; Sánchez et al. 2019; Pomares et al. 2018; Kragh et al. 2015; Martínez et al. 2020). However, the process of manually classifying actual 3D data obtained from ground vehicles traversing natural terrains presents considerable difficulties in terms of both labor and accuracy. Addressing this issue, Sanchez et al. (2019)

employed synthetic 3D laser scans generated by Gazebo, automatically labeled point by point. Martínez et al. (Martínez-Rodríguez et al. 2020) did similar work and subsequent comparative analyses of seven supervised learning techniques RF and gradient boosted trees (GBT) classifiers as top performers in terms of accuracy indices. Another comparative analysis by Pomares et al. (2018) included 23 state-of-the-art machine learning methods using the “Classification Learner” App (Statistics and Machine Learning Toolbox—mathworks.com 2025). Notably, techniques such as decision trees, discriminant analysis, and logistic regression demonstrated excellent performance in terms of training and prediction times, as well as prediction accuracy.

Several studies have explored the effectiveness of SVMs employing various kernels and descriptors, consistently achieving high accuracy (McDaniel et al. 2010; Kragh et al. 2015; Ahtiainen et al. 2017; Bellone et al. 2018). Navarro et al. (Santamaria-Navarro et al. 2015) found that Gaussian Process classification outperforms SVM and naive parametric classification with just two features, local roughness and slope, while also offering significantly reduced computational load. Moreover, utilizing only five features is found to be sufficient by Kragh et al. (2015) for achieving high accuracy in classification tasks, reducing computational complexity. In efforts to reduce the dependency on manually labeled training data, self- and semi-supervised learning approaches have gained traction (Reina et al. 2015, 2016; Suger et al. 2015). Reina et al. (2016) proposed the integration of two self-learning classifiers—one based on LiDAR data and the other based on stereo data. By statistically combining the outputs from these single-sensor classifiers, their individual strengths are harnessed to achieve superior overall performance compared to their standalone usage.

Furthermore, Markov Random Fields (Guo et al. 2011; Anguelov et al. 2005; Wolf et al. 2005) and Bayesian Generalized Kernel Inference (Shan et al. 2018) have been explored in the literature. Lalonde et al. (Lalonde et al. 2006) developed a 3D data segmentation approach using Gaussian mixture models and Bayesian classification to categorize objects into “scatter,” “linear,” and “surface” classes. Vu et al. (2017) proposed a segmentation method that integrates gradient thresholding, adaptive breakpoint detection, and mean height evaluation, yielding improved segmentation outcomes compared to the naive gradient field approach, particularly in complex areas with similar gradient values near boundaries or obstacles.

Various terrain descriptors have been explored to enhance terrain classification and assessment performance. For instance, Bellone et al. (2014) introduced the unevenness point descriptor (UPD) for terrain analysis, which uses principal component analysis (PCA) to estimate normal vectors of query points. UPD is derived from these normal vectors, enabling the system to analyze local surface geometry and identify non-traversable obstacles or highly irregular ground. Additionally, Meng et al. (2018) proposed a novel terrain description method where the slope angle is computed using RANSAC and least squares. Vu et al. (2017) proposed a segmentation method that integrates gradient thresholding, adaptive breakpoint detection, and mean height evaluation, yielding improved segmentation outcomes compared to the naive gradient field approach, particularly in

complex areas with similar gradient values near boundaries or obstacles. A study by Darganzy et al. (Dargazany and Berns 2014) introduced pixel-based surface normals derived from a stereo camera, for environment perception. These surface normals were utilized for terrain point cloud segmentation and classification based on traversability criteria. Building upon this study, the author in (Dargazany 2019) proposed a method to detect all surfaces in the 3D reconstructed terrain as superpixel surfaces using image segmentation techniques. Their superpixel surface traversability analysis (SSTA) approach classified terrain surfaces into five main categories: traversable, semi-traversable, non-traversable, unknown, and undecided.

3.2.2 | Deep Learning in Geometry-Based Terrain Classification

Deep learning-based geometry-driven terrain classification has gained prominence due to the unique challenges posed by off-road environments, where the absence of structured road features makes conventional 3D point cloud segmentation techniques ineffective. Unlike structured roads, off-road traversability lacks clear lane markings and consistent boundaries, requiring specialized learning approaches such as weakly supervised learning and transfer learning. Gao et al. (2019) addressed this challenge by introducing a semi-supervised method for off-road drivable area extraction using 3D LiDAR data. Their framework specifically targeted the ambiguity of the gray zone—regions that are technically drivable but not ideal—by designing a dual-branch VGG-based network that learns two classification surfaces to separate drivable and obstacle zones. This probabilistic approach significantly reduces reliance on human-annotated labels, demonstrating superior performance to fully supervised methods while leveraging automatic labeling from vehicle driving data. Similarly, Zhou et al. (2023) proposed MAPC-Net, a segmentation algorithm that integrates a gated fusion module for combining Point-Tensor and Cylinder representations, alongside a multi-layer receptive field module to improve feature extraction for objects of varying scales. Their approach, trained on an off-road data set generated in the CARLA simulator and enhanced with linear interpolation, outperformed traditional projection-based and voxel-based segmentation methods, such as SalsaNext (Cortinhal et al. 2020) and PVCNN (Liu et al. 2019).

Graph-based and projection-based methods have been explored to enhance off-road semantic segmentation by leveraging geometric structures and efficient feature aggregation. Liu et al. (2024a) proposed a hybrid 3D segmentation model that integrates graph convolution to aggregate local geometric features with a projection module comprising 2D projection, FCN, and a grid-to-point block. This combination captures spatial relationships while mitigating the class imbalance inherent in cross-country datasets through an improved loss function. On the other hand, Schneider and Stemmer (2024) incorporated multi-view representations and deep learning, and proposed a multi-task CNN for off-road segmentation using a bird's eye view (BEV) representation of 3D LiDAR data. Their approach integrates a keypoint feature pyramid network with an FCN using a ResNet-50 encoder backbone, enabling both semantic segmentation and 3D object detection. The system was

developed within the “Gazebo” Simulator (Koenig and Howard 2004) and “ROS2” (Open Robotics 2022) framework, for data set generation and real-time semantic mapping. The results demonstrated strong performance in simulation, though real-world deployment showed limitations due to domain gap issues. Shan et al. (2023) present a segmentation-modeling-refinement approach that synergizes deep learning with Gaussian process (GP)-based modeling for off-road terrain analysis. The framework first segments road surfaces, obstacles, and vegetation using a neural network, followed by an improved GP-based method to refine the road surface model while excluding outlier point clouds. A composite filter is then applied to enhance obstacle segmentation by filtering misclassified points. Field experiments validate the effectiveness of this hybrid methodology, demonstrating precision and recall improvements of up to 14% and 9%, respectively, over state-of-the-art approaches. Dabbiru et al. (Dabbiru et al. 2020) used simulated labeled LiDAR point cloud data for off-road navigation. It trains a CNN-based segmentation model (SqueezeSeg (Wu et al. 2018)) to identify obstacles and terrain features. Evaluation at 8-beam and 16-beam LiDAR resolutions shows a 5% accuracy boost in obstacle detection with 16-beam LiDAR, while 8-beam performs similarly for vegetation and trails.

3.3 | Hybrid Approaches

In numerous literature focused on terrain perception, there is a general trend of incorporating multiple sensor modalities and feature types for improved outcomes. Recently, multiple studies have concentrated on merging data from diverse sensor modalities and extracting distinct types of features to take advantage of their complementary capabilities. Integrating sensors within the same modality has demonstrated improvements in classification performance. Csik et al. (2021) utilized both accelerometer and gyroscope data to classify six different terrains, finding that significantly higher recognition rates were achieved with gyroscope signals compared to using only accelerometer data. Similarly, Otte et al. (2016) showed that including gyroscope data with accelerometer data increased accuracy for all networks by 2%–3%. Dutta and Dasgupta (2017) employed non-vision-based sensors (GPS, IMU, Laser, and Metal Detector) with ensemble learning to classify five terrain types. Their results show that this method, using multiple weak classifiers, outperforms traditional SVM approaches while being lightweight and efficient, reducing computational resource requirements for mobile robots.

The utilization of hybrid methodologies could improve the system's potential to adapt to various terrains, hence guaranteeing reliable performance in real-world scenarios. For example, Halatci et al. (2007) used a multimodal method for terrain classification that included color, texture, range, and vibration information. This involved training a maximum likelihood classifier using a mixture of Gaussian models (MoG) and an SVM classifier. The classifiers' results were then combined using Bayesian fusion techniques. The study showed that using classifier fusion approaches considerably improved overall performance. The fusion approaches can be broadly classified into two groups, depending on the characteristics of the combined data. The first category includes a combination of

data acquired from proprioceptive and exteroceptive sensors, such as inertial sensors and LiDAR or camera systems. By integrating vehicle dynamics measurements with environmental observations, these approaches provide a comprehensive view of terrain characteristics and navigational challenges. The second fusion technique combines geometric and appearance-based data obtained from sensor data or image representations.

3.3.1 | Integration of Proprioceptive and Exteroceptive Data

In the area of proprioceptive and exteroceptive method fusion, a handful of papers have explored the use of vibration and imagery together to classify terrains. Brooks and Iagnemma (2012), utilized a proprioceptive terrain classifier to distinguish terrain classes based on features derived from rover-terrain interaction. The labels from this classifier were then used to train an exteroceptive (i.e., vision-based) terrain classifier. SVM with hand-labeled data was used as the proprioceptive classifier, whereas the exteroceptive classifier was self-supervised. In a separate study conducted by Otsu et al. (2016), the concept of co-training two classifiers for vision and vibration data is introduced. These classifiers are then iteratively re-trained using each other's output.

The difference from the previously mentioned paper lies in the fact that the training methods described in Brooks and Iagnemma (2012) are one-way, meaning that vibration is employed to train the vision-based classifier, but not the other way around. The suggested co- and self-training methods effectively merge the vision- and vibration-based classifiers, leading to a 7.7% decrease in errors compared to the naive supervised classifier, even when using a very small amount of training data.

In a similar context, Kurup et al. (2020) conducted an analogous experiment where images were manually labeled and aligned with vibration data from the IMU to produce a fused feature vector for training the SVM. Additionally, Wang et al. (2023) integrated vibration and imagery data, employing principal component analysis (PCA) for dimensionality reduction. Simultaneously, texture features from the terrain images were extracted using the gray-level co-occurrence matrix (GLCM) technique. An enhanced weighted K-nearest neighbor (WKNN) algorithm was utilized for terrain classification. Their findings demonstrate a noteworthy improvement in classification accuracy with the proposed hybrid method at two different vehicle speeds compared to single-modal approaches. Reina et al. (2018) proposed a multi-class terrain classification approach using SVM. Their two-step process first assigns vision-based features to a terrain patch and later refines classification using proprioceptive features when the patch is traversed. This fusion improves accuracy, overcoming lighting sensitivity issues in vision-based methods.

The fusion of audio and vision for terrain classification has been explored in a couple of studies (Zürn et al. 2020; Kurobe et al. 2021). Zürn et al. (2020) proposed a framework for terrain classification that uses an unsupervised proprioceptive classifier that learns from vehicle-terrain interaction sounds to

self-supervise an exteroceptive classifier, enabling pixel-wise semantic segmentation of images. They employed Short-Time Fourier Transform (STFT) to generate spectrogram representations of the audio signal. A notable advantage of their self-supervised approach is its scalability with the duration of experimental runs and its independence from manual labeling of terrain transitions. Firkat et al. (2023) introduced FGSeg, which processes point cloud data by first performing coarse ground segmentation to classify points into candidate, obstacle, and coarse ground seed points. It then applies multi-region ground refinement to classify terrain types within the coarse seed ground and uses an adaptive height threshold to merge the ground points to refine the segmentation. Most of the recent works in proprioceptive-exteroceptive hybrid methods are situated within the domain of self-supervised learning. These approaches address key challenges, such as the dependency on manual annotations and the inflexible definitions of traversability, which limit the scalability across different vehicles and environments. Self-supervised methods, by leveraging actual driving experiences, can learn traversability directly from sensor data without requiring human-provided labels. However, these methods often face uncertainty issues, particularly in non-traversed regions, which can lead to unreliable navigation (Schmid et al. 2022). Wellhausen et al. (2019) propose a self-supervised framework that combines proprioceptive sensor readings from six-axis force-torque (F/T) sensors and exteroceptive imagery to predict terrain properties. Their method projects foothold positions onto RGB images, allowing the CNN, based on ERFNet (Romera et al. 2017), to learn pixel-accurate terrain property predictions. A novel ground reaction score, calculated using principal component analysis (PCA) of force-torque data, quantifies terrain difficulty for robot navigation. Seo et al. (2023a) incorporates both vehicle-specific and terrain data. They utilize LiDAR-based SLAM to track wheel-terrain interactions and model traversability from 3D point clouds using a Positive Unlabeled (PU) learning method. Their framework predicts both point-wise traversability and binary classification of traversable terrain, addressing uncertainty in non-traversed regions. With RandLA-Net (Hu et al. 2020) as the backbone, their system enables real-time traversability estimation without human intervention. One of the earlier works in this paradigm is by Dahlkamp et al. (2006), combining a laser range finder with a camera to create a drivability map. The laser identifies drivable areas, which serve as training data for the vision algorithm. The system then uses a Mixture of Gaussians (MOG) model with k-means clustering to classify road patches and generate a visual model of the road. The approach was validated in the 2005 DARPA Grand Challenge. Castro et al. (2023) introduced a method combining IMU, camera, and LiDAR to predict traversability costmaps. They generate pseudo-ground truth from IMU power spectral density and incorporate robot velocity into the model to improve performance on rough terrain at high speeds. Sathyamoorthy et al. (2022) proposed a self-supervised approach using RGB images and robot velocities along with IMU vibrations and odometry errors to predict surface navigability. Their method extends the Dynamic Window Approach (DWA) (Fox et al. 1997) for outdoor navigation, accounting for surface costs. The proposed framework, 'TerraPN', is illustrated and described in Figure 6. Kahn et al. (2021) developed a predictive model, BADGR, that autonomously labels events like collisions and bumpiness from

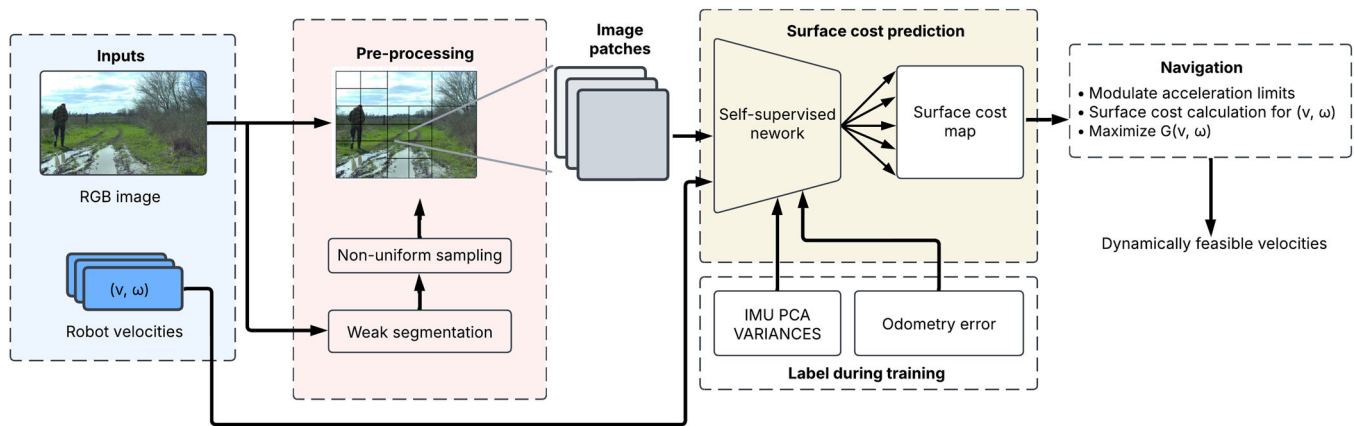


FIGURE 6 | TerraPN architecture proposed by (Sathyamoorthy et al. 2022): The method processes RGB images and robot velocities to generate a surface cost map. A weak segmentation technique differentiates regions by pixel intensity, followed by nonuniform patch sampling. Extracted patches are fed into a self-supervised network to predict a 4×1 label vector, whose weighted norm determines the surface cost. The cost map is constructed by arranging cost-labeled patches spatially, with blue indicating low and yellow high costs. A navigation algorithm then computes dynamically feasible velocities with minimal surface costs in real time, iteratively updating per input image and velocity vector.

real-world data. They use this labeled data to train a neural network that predicts future events based on camera and action inputs. Karnan et al. (2023) introduced STERLING, a self-supervised terrain representation learning approach using multi-modal data from RGB cameras, odometry, IMU, and proprioceptive/tactile sensors. It leverages power spectral density (PSD) representations and computes a correlation objective between visual and inertial-proprio-tactile signals for learning terrain features. Frey et al. (2023) developed Wild Visual Navigation (WVN), a self-supervised system for traversability estimation using RGB images that uses Vision Transformers (ViT) with the DINO method and derives a continuous traversability score based on discrepancies between the robot's velocity and reference commands. Jung et al. (2024) combined stereo-depth information and projected vehicle trajectories with class-agnostic semantic masks from the segment anything model (SAM) (Kirillov et al. 2023). By using a pre-trained image encoder, a traversability decoder, and contrastive losses, the method improves traversability learning with human driving data and mask proposals. Their proposed framework, “VSTRONG,” is illustrated and described in Figure 7. Bajracharya et al. (2009) proposed a hybrid approach in the LAGR program, combining short-range geometry-based classifiers with long-range image-based classifiers.

The synergy between proprioceptive and exteroceptive sensors not only enhances the robustness of the models but also reduces the need for expensive labeled data. As this field progresses, future research should focus on further optimizing the integration of these modalities to handle more complex terrains and varying environmental conditions.

3.3.2 | Integration of Geometry and Appearance Features

Terrain classification, methods based on geometry and appearance, often leverage data from LiDAR and cameras. Zhou

et al. (2012) conducted a study where they utilized a hybrid LiDAR/camera-based method. This method adopted a self-supervised learning model, where a visual classifier was trained using the outcomes of a LiDAR-based ground plane identification technique. An ideal feature set was chosen by employing AdaBoost and the Fuzzy SVM method. After that, SVM was used with the selected features to find a decision hyperplane in the feature space for the visual classifier. Maier et al. (2011) used a 2D laser sensor attached to a robot to generate labels, facilitating the self-supervised classification of corresponding images. Wellhausen et al. (2020) highlighted the effectiveness of a feature extractor and normalizing flow model utilizing RGB, depth, and surface normals, achieving over 95% ROC curve area and robustness to out-of-distribution samples. Both studies, (Ono et al. 2015) and (Bradley et al. 2015), used a random forest classifier-based approach. Ono et al. (2015) employed geometric features derived from stereo pairs of images and appearance-based features, including gray intensity and image gradients, for terrain classification. Subsequent analysis in Bradley et al. (2015) emphasized the importance of adapting algorithms to varying environmental conditions and highlighted performance differences between seasons, stressing the need for robust and adaptable scene understanding systems for off-road navigation. Dai et al. (2014) utilized a 3D laser sensor and monocular vision sensor with self-supervised learning to adaptively classify terrain types based on features extracted from the environment. In another study, Laible et al. (2013) proposed a multimodal technique that used Conditional Random Fields (CRF) and compared it with Markov random field (MRF)-based methods. Unlike MRFs, which model the joint probability of labels and features, CRFs model the conditional probability of labels given the features, making them discriminative. The authors found that, whereas MRFs have shown effectiveness in a variety of tasks, including terrain classification, CRFs generate significantly better classification results in outdoor terrain classification environments. Milella et al. (2014) introduced a Radar-Vision Combination method, leveraging radar measurements to identify ground patches in camera images. Subsequently, visual

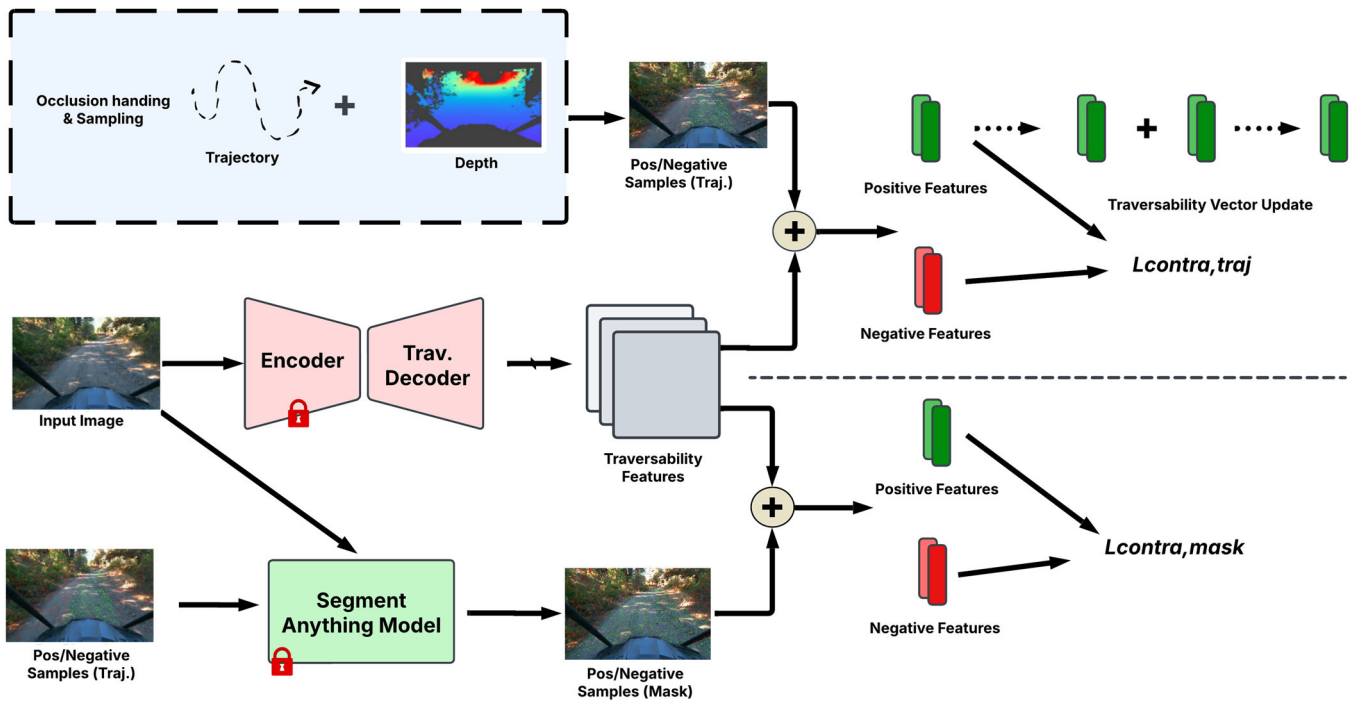


FIGURE 7 | V-STRONG method proposed by (Jung et al. 2024): Stereo-depth filtering removes occluded trajectory points before projecting the trajectory into image space. Positive and negative points are sampled using the trajectory and SAM-predicted masks. A pre-trained image encoder and a traversability decoder extract features, with contrastive losses applied for training. The traversability vector is iteratively updated using a running average of positive features, enabling similarity-based traversability estimation at test time. Dashed gray arrows indicate gradient stops, with the encoder remaining frozen during training.

segmentation of the entire scene was conducted using a one-class classification system employing a multivariate Gaussian model of the terrain and a Mahalanobis distance-based outlier detection approach. The visual classifier trained across the region where radar overlaps with the camera field of view in a self-supervised manner, avoids the need for manual labeling and enables real-time implementation.

Seo et al. (2023b) used RGB images and LiDAR data to generate traversability labels, employing Positive-unlabeled (PU) learning to tackle the challenge of unlabeled negative samples. They use a 2D Normalizing Flow model and self-supervised representation learning to distinguish traversable from non-traversable regions. However, the method's reliance on low-level features from RGB and LiDAR could limit its ability to capture complex global scene information. Wellhausen et al. (Wellhausen et al. 2020) explore anomaly detection using RGB-D data, demonstrating that incorporating geometric information (depth) significantly outperforms RGB-only inputs for traversability assessment. Their work shows how anomaly detection methods, particularly Real-NVP (Dinh et al. 2016), outperform traditional autoencoders and Deep-SVDD (Ruff et al. 2018) models. Schmid et al. (Schmid et al. 2022) extend the idea of self-supervised learning by incorporating RGB stereo and LiDAR data to project wheel positions into 2D images for traversability estimation. This approach leverages autoencoders to identify high- and low-risk terrains based on reconstruction errors. The reliance on reconstruction error could also lead to misclassifications, especially when the terrain differs significantly from previous experiences. Jeon et al. (Jeon et al. 2024) utilized RGB-D data combined with a fusion network (ResNet

blocks) and a footprint supervision module for traversability estimation. Their method was validated against WayFAST (Gasparino et al. 2022), BEVNet (Shaban et al. 2022), and GA-Nav (Guan et al. 2022), with results demonstrating performance comparable to the supervised method. Zhou et al. (Zhou et al. 2012) apply a two-stage process for ground surface detection using LiDAR data, training a visual classifier to differentiate between ground and non-ground regions. While their method adapts to environmental changes and improves classifier performance through online learning, it is susceptible to errors caused by incorrect training data, which could degrade performance. The introduction of morphological operations and the use of AdaBoost for feature selection help mitigate this, but more robust data-cleaning methods could enhance reliability. Chen et al. (2024) introduced the ALTER framework, which adapts visual models online using LiDAR data to estimate traversability in novel environments. Their approach is effective in handling long-range visual observations and LiDAR data, ensuring robustness in previously unseen environments.

While these self-supervised approaches demonstrate promising advancements in terrain traversability learning, challenges remain in terms of scalability, adaptability, and the handling of diverse, dynamic environments. Many methods rely on a combination of visual and geometric data, but their effectiveness is often constrained by the complexity of terrain and the limitations of specific sensors. Further refinement in data fusion techniques and model generalization could enhance the robustness and scalability of these systems. Table 4 summarizes key hybrid approaches in off-road terrain classification.

TABLE 4 | Overview of hybrid methods used for terrain classification.

Reference	Method	Data	Terrain type
Seo et al. (2023b)	PSPNet with ResNet50+normalizing flow	RGB and LiDAR	Drivable and non-drivable
Wellhausen et al. (2019)	Self-supervised learning with ERFNet	RGB-D and six-axis force-torque (F/T) sensors	Asphalt, gravel path, grass, dirt, sand
Zürn et al. (2020)	Deeplabv3+/AdapNet++/ENet/BiSeNet, Fast-SCNN with ResNet50/DRN-D-105/Xception/MobileNet-V2/EfficientNet/Custom/Xception/Custom	RGB and microphone	Grass, gravel, asphalt, cobblestone, parking lot
Schmid et al. (2022)	ResNet18,50 with wheel positions projected into 2D images	RGB, Stereo, and LIDAR	Drivable and non-drivable
Jeon et al. (2024)	ResNet blocks + footprint supervision module	RGB-D	Drivable and non-drivable
Sathyamoorthy et al. (2022)	Dynamic Window Approach (DWA) with PCA on IMU data	RGB, IMU, and LiDAR	Concrete, grass, asphalt, rocks
Jung et al. (2024)	Contrastive learning + semantic masks from SAM	RGB-D	Drivable and non-drivable
Karnan et al. (2023)	Self-supervision, + correlation between Visual Encoder and Inertial-Proprio-Tactile Encoder	RGB, odometry, IMU, and tactile sensors	Marble rocks, yellow bricks, grass, cement sidewalk, pebbled sidewalk, red bricks, bush
Frey et al. (2023)	DINO + MLP head	RGB, pose, and velocity measurements.	Hilly, forest, grass
Chen et al. (2024)	ALTER with adaptive model selection	RGB and LiDAR	Low, medium, and high cost

4 | Critical Analysis of Terrain Classification Methods

The complexities of off-road environments make it challenging to develop a one-size-fits-all solution. Variability in terrain composition, surface textures, and lighting conditions can distort sensor measurements, leading to inaccuracies in classification outcomes. Moreover, the dynamic nature of off-road environments, including the presence of moving obstacles and changing topography, adds to the complexity of the classification task. Addressing these challenges requires a holistic understanding of the interplay between sensor modalities, feature extraction techniques, and machine learning algorithms. One of the key challenges in terrain classification research stems from the lack of well-annotated datasets tailored to off-road environments and diverse environmental conditions encountered in real-world scenarios. Studies utilize a variety of sensor modalities, ranging from LiDAR and cameras to inertial sensors and GPS, each offering unique insights into the surrounding terrain. Furthermore, variations in terrain types, such as rocky terrain, forest trails, or desert landscapes, necessitate adaptable classification strategies capable of generalizing across diverse conditions.

4.1 | Overview of the Comparative Studies

Several papers examining comparative studies across various methods and sensing modalities have been identified. These

studies involve testing multiple algorithms or techniques on common datasets or under similar conditions to assess their effectiveness in real-world scenarios. By comparing the results obtained from various approaches, researchers can identify the strengths and weaknesses of each method and make informed decisions about which techniques are most suitable for this specific applications. Table 5 provides a concise summary of findings from these comparative studies, facilitating a comprehensive understanding of the comparative landscape in terrain classification research.

Exteroceptive terrain classification studies are highly dominated by CNN architectures. Sai et al. (2024) and Vishnyakov et al. (Vishnyakov et al. 2020) investigated a variety of encoder-decoder designs for the purpose of natural terrain classification. The first one implemented seven transfer learning models that were trained on ImageNet, while the latter employed a simulated data set in Unreal Engine 4 and Cityscapes for pre-training. Their study subsequently refined the models on their original off-road data set, achieving a 2.7% increase in results using ResNet34 and DeepLabV3 in comparison to Cityscapes pre-training. Clark et al. (2019) investigated how different CNN input types affect terrain classification accuracy. Findings indicate that although RGB input type shows better validation accuracy, color tends to overfit training data, suggesting exploration of grayscale and disparity maps in certain cases. Singh et al. (2022) applied transfer learning with Hyperspectral Images (HSI), comparing CNNs and three MLP models. They

TABLE 5 | Summary of comparative and validation studies.

References	Category	Comparison focus	Data set information	Comparison results
Peng et al. (2024)	Exteroceptive	DeepLabV3+, DaNet, PSPNet, LMFFNet, PP-LiteSeg, PIDNet-S	WVD Data set (Zou et al. 2023) RELIS (Jiang et al. 2021) RUGD (Wigness et al. 2019)	Proposed LB-ASN method outperformed other approaches with lower params, lower FLOPs, and lower required memory.
Liu et al. (2024b)	Exteroceptive	Segformer-b1, GANav, FastSCNN, PSPNet Deeplabv3, GCNet ANNet, AR-Net	RELLIS (Jiang et al. 2021) RUGD (Wigness et al. 2019)	The proposed AR-Net outperformed other approaches with the lower MParams, GFLOPs, and highest FPS.
Jeon et al. (2024)	Hybrid	WayFAST BEVNet, GANav, Agishev et al. (Agishev et al. 2022), author proposed network	RELLIS (Jiang et al. 2021) ORFD (Min et al. 2022)	Author proposed method outperforms others in path generation, freespace detection, and real-time feasibility
Chen et al. (2024)	Hybrid	GANav (Pretrained on RELIS) GANav (Pretrained on RUGD) Adaptive SVM (Zhou et al. 2012)	10s buffers, each with 10 training and 4 validation frames	ALTER outperforms baselines with model selection significantly improving reactivity to new environments, as demonstrated by higher mIoU in the first 15 s compared to ALTER without model selection.
Sai et al. (2024)	Exteroceptive	VGG16, DenseNet-121, MobileNet, Inception V3, ResNet-101, DenseNet-201, and ResNet-50 trained on ImageNet	30,000 images in total for 4 classes	MobileNet achieved the highest accuracy of 90%.
Vishnyakov et al. (2020)	Exteroceptive	Encoder: U-Net, DeeplabV3 Backbone: ResNet18, ResNet34, MobileNetV2, ShuffleNetV2, EfficientNet-B0	Over 100,000 images and LiDAR scans	ResNet34 + DeeplabV3 achieved the highest performance mIoU(85.2%).
Clark et al. (2019)	Exteroceptive	Different CNN input formats - RGB, Stack-RGB, Grayscale, Stack-Gray, Disp. maps, Stack-RGB+disp, Stack-Gray+disp.	4688, 3866, and 3218 images for high, medium, and low mobility terrains, respectively, totaling 11,772 images.	RGB input performed best due to color information aiding validation accuracy, but over-reliance on color led to poor testing accuracy.
Singh et al. (2022)	Exteroceptive	CNN and MLP on hyperspectral image (HSI)	Indian Pines (IP): 145 x 145 image, 224 spectral bands (400–2500 nm) University of Pavia (UP): 610 x 610 image, 103 spectral bands (430–860 nm)	The MLP model with four layers achieved superior performance with a test accuracy of 99.95% on the IP data set and 98.71% on the UP data set.

concluded that choosing between CNN and MLP transfer learning depends on the application and resources. For agricultural research emphasizing spatial features, CNNs are preferred. Conversely, in military remote sensing focusing on spectral features over spatial, MLPs are favored.

The outcomes of comparative studies can vary significantly depending on factors such as the complexity of the terrain, the quality of the sensor data, and the feature extraction techniques employed. In some cases, certain methods may outperform others in terms of classification accuracy, while in other scenarios, the differences between methods may be less pronounced. Additionally, comparative studies often reveal insights into the robustness and generalizability of different approaches across diverse environmental conditions.

4.2 | Strengths and Trade-Offs of Proprioceptive and Exteroceptive Sensors

A comparison of proprioceptive and exteroceptive classifiers demonstrates each approach's unique strengths and limitations (Krebs et al. 2009). While both demonstrate commendable performance under certain settings, inherent constraints underscore the need for a comprehensive understanding. Proprioceptive techniques mandate the vehicle to physically traverse the terrain for data collection and classification of terrain, which may cause damage or wear. Depending on the experimental platform or vehicle, the proprioceptive data of interest (e.g., accelerometer and gyroscope) and the corresponding references can vary significantly even within the same operating state and environment. Some studies, such as those by Bai et al. (2019), Brooks and Iagnemma (2005), and Dupont et al. (2008), have made efforts to utilize two experimental platforms. However, to generalize experimental results for real-world applications, analysis must be conducted across a range of platform types. Vehicle speed also greatly affects proprioceptive classification performance. Moreover, proprioceptive sensors often have limited sensing ranges, restricting coverage in large areas. Sensor noise and uncertainty, including calibration errors and environmental impacts, present additional challenges (Yuksel et al. 2010; Quinchia et al. 2013). Proprioceptive sensors also struggle to reliably detect certain terrain features, such as fine textures or minor elevation changes, especially in complex environments.

On the other hand, exteroceptive sensors exhibit poor performance in challenging environments or adverse weather conditions. They are highly dependent on environmental factors like illumination, dust, snow, and so on. (Linnhoff et al. 2022; Anand et al. 2022; Rodríguez-Rodríguez et al. 2024). To address these challenges, a large amount of training data is required across various environmental settings, which can be both time-consuming and costly. A solution for this is using synthetic data from commercial games and simulations to create environmental setups and gather training data along with ground truth. For example, Zhang et al. (Zhang et al. 2021) utilized Grand Theft Auto V (GTAV) and RenderDoc (Renderdoc 2018) to simulate foggy and dusty images from the same scene. Processing the complex data generated by exteroceptive sensors can be computationally intensive and time-

consuming, posing challenges for real-time applications. Moreover, occlusion and interference from obstacles or other vehicles can lead to incomplete or distorted sensor readings.

Proprioceptive sensors provide real-time terrain contact information, which makes them a promising avenue for self-supervising exteroceptive classifiers. In scenarios where annotated data is entirely unavailable, leveraging proprioceptive data to guide the learning process of exteroceptive sensors presents an effective solution. Furthermore, in resource-constrained environments where real-time processing of point clouds or depth information is infeasible, vision sensors can be supervised by proprioceptive sensors to achieve comparable classification accuracy. A good level of balance must be found when fusing proprioceptive and exteroceptive sensors, particularly in how the input data is processed and how the algorithms are trained. Achieving an effective combination of sensor outputs requires careful consideration to ensure optimal performance without overwhelming computational resources. Therefore, more research should focus on developing lightweight yet efficient frameworks that can seamlessly handle and integrate diverse sensor data, enabling real-time processing and accurate terrain classification in off-road environments.

4.3 | Geometry-Based Versus Appearance-Based Methods

Geometry-based terrain classification is predominantly driven by LiDAR technology, which creates detailed 3D maps of the environment. LiDAR point cloud data is susceptible to noise, influenced by the vehicle's speed and direction. Also, LiDAR is generally costly and more energy-consuming. According to Wang et al. (2021) there are two potential trends to address this: reducing the cost by producing LiDAR for specific tasks with limited detection distances, and using solid-state LiDAR to lower costs. Geometry-based terrain classification algorithms have accuracy and robustness issues with feature extraction. Sensor noise and calibration issues can cause geometric measurement errors and terrain feature misclassification. Due to their reliance on an accurate geometric model, these approaches may struggle to manage complicated terrains with irregular surfaces, occlusions, or variable elevation profiles.

Appearance-based terrain classification algorithms have environmental unpredictability and sensor-specific constraints. These approaches are sensitive to lighting, weather, and surface texture, which can alter terrain feature appearance and classification accuracy. Appearance-based approaches may also need a lot of computational power for image processing and feature extraction, rendering them unsuitable for real-time or resource-constrained applications.

Despite these drawbacks, geometric-based solutions have advantages over appearance-based approaches in some cases. Their capacity to capture ground surface geometric features, including slope, curvature, and roughness, is useful for terrain characterization and traversal analysis. Geometric aspects are easier to grasp and understand, making landscape models and dynamics more relevant.

In contrast, appearance-based approaches capture surface texture and contextual information from visual sensor data well. These approaches may distinguish terrain types and detect surface property variations using rich visual signals, including color, texture, and shape. If visual appearance is important, appearance-based techniques are ideal for landscape classification tasks including detecting flora, water bodies, and man-made objects.

4.4 | Merits and Challenges of the Hybrid Method

The fusion of appearance-based and geometry-based methods in terrain classification, along with the integration of proprioceptive and exteroceptive sensors, combines the strengths of multiple sensing modalities, offering a comprehensive understanding of the terrain. Appearance-based techniques excel in capturing surface texture and visual cues, while geometry-based methods provide precise geometric representations, facilitating accurate terrain detection and elevation estimation. By integrating texture and geometric features, along with proprioceptive and exteroceptive sensor data, hybrid methods can effectively discriminate between different terrain types and obstacles, leading to robust navigation performance. Wang et al. (2021) compared the pros and cons of LiDAR and camera sensors for environmental perception in autonomous vehicles, highlighting the critical role of sensor fusion in advancing autonomous vehicle technology.

Hybrid methods of terrain classification, leveraging multi-sensor fusion technologies, offer comprehensive advantages in autonomous systems, from humanoid robots to AVs. The analogous idea of the hybrid method is data maximization, which essentially means that the more data collected, the smarter an autonomous system could get (Keep or Not to Keep...Data for Self-Driving Vehicles?—drivesweden.net 2025). In 2016, Intel estimated that each autonomous vehicle would generate 4000 gigabytes of data per day, a volume that would cost about \$350,000 to store for a year at Amazon's current prices (Intel Editorial: For Self-Driving Cars, There's Big Meaning Behind One Big Number: 4 Terabytes—businesswire.com 2025). Nearly a decade later, as the autonomous sector operates under tighter budgets (Condé Nast 2025), Yeong et al. (Yeong et al. 2021) suggested that combining reinforcement learning (RL) techniques with supervised learning algorithms could help to reduce computational power, training data requirements, and training time.

RL is a machine learning method that uses feedback from actions and experiences to train models in an interactive environment, while supervised learning utilizes labeled data for training. However, the performance of ML/DL models for terrain classification is heavily influenced by the quality of the data set used. As per a study by Hao et al. (Error-riddled data sets are warping our sense of how good AI really is—technologyreview.com 2025) in MIT Technology Review, the ImageNet data set (Deng et al. 2009) exhibits an estimated label error rate of 5.8%. In contrast, the test set for QuickDraw (GitHub - googlecreativelab/quickdraw-data set: Documentation on how to access and use the Quick, Draw! Data set. —github.com 2025), which comprises hand-drawn images, shows a

higher estimated error rate of 10.1%. Poor data quality can lead to the proverbial “garbage-in, garbage-out” scenario. DL systems are susceptible to adversarial attacks, where slight alterations to input data can lead to significant misclassifications (Ren et al. 2020; Ma et al. 2021). Missing values and calibration problems in the data, overfitting, biases in the gathered datasets, imprecision and converting multi-sensor data into a single frame of reference are also a few of the challenges associated with the learning-based hybrid methods.

5 | Overview of Available Off-Road Datasets

AGV perception tasks like terrain classification necessitate quality datasets. These datasets allow researchers to benchmark algorithms and assess various tasks in a variety of environmental conditions. Such benchmarking is essential for the development of algorithms that are both reliable and robust and are capable of managing the complicated nature of off-road environments. Urban datasets such as Apolloscape (Huang et al. 2018), CityScapes (Cordts et al. 2016), and KITTI (Geiger et al. 2013), which contain hundreds of thousands of annotated urban images, are highly popular and extensively used in research. Despite this, the research community is lacking high-quality off-road datasets. To bridge this gap, researchers have collected and circulated data from both real-world autonomous operations and simulated synthetic scenarios.

5.1 | Real-World Off-Road Datasets

Real-world off-road datasets are normally captured by autonomous platforms, such as robotic vehicles, all-terrain vehicles (ATVs), or drones equipped with various sensors, including cameras, stereo pairs, and LiDAR systems. Cameras and stereo pairs provide visual data, while LiDAR systems generate point cloud data that captures the three-dimensional structure of the environment. Annotation labels terrain types and components of the data set after collection. Common annotation tools include LabelMe (Russell et al. 2008), VGG Image Annotator (VIA) (Dutta and Zisserman 2019), and CVAT (CVAT—cvat.ai 2025). Annotated datasets serve as the foundation for training machine learning algorithms, enabling them to learn from labeled examples and generalize to new, unseen data.

Among all the available real-life datasets, the RELIS-3D data set (Jiang et al. 2021) stands out as the largest and most popular for off-road terrain classification. It was collected on the Relis Campus of Texas A&M University and includes annotations for 13,556 LiDAR scans and 6235 images, covering 20 different classes. This data set is highly valuable for researchers due to its comprehensive documentation and diverse environment. However, class imbalance poses a challenge, requiring the application of bias based on class frequency during training. Similarly, the RUGD data set (Wigness et al. 2019), containing 7436 images for 24 different classes, also exhibits inhomogeneous distribution. Guan et al. (Guan et al. 2022) benchmarked both datasets for semantic terrain segmentation using GA-Nav, a novel group-wise attention mechanism, which improved performance (in terms of mIoU) by 6.67%–10.85% on RUGD and 4.18%–10.31% on RELIS-

3D, regardless of the backbone choice. The Freiburg Forest (Valada et al. 2017) data set extended the focus from RGB-only datasets to include RGB, Near-Infrared (NIR), and depth data. This study evaluates segmentation across a variety of spectra and modalities, identifying the best performers, and fuses them using various deep convolutional neural network (DCNN) architectures. Results show that fusing multiple spectra, especially NIR with RGB, enhances segmentation accuracy in forested environments. Both the Yamaha-CMU Off-Road Data set (YCOR) (Maturana et al. 2017) and the CaT (Sharma et al. 2022) data set are obtained by utilizing ATVs. The YCOR data set covers three distinct seasons, including snowy trails, and the CaT data set classifies roads into three categories based on the capacity of vehicles to traverse them: sedan, truck, and off-road. The GOOSE data set (Mortimer et al. 2023) incorporates the highest number of semantic classes, totaling 64, to segment RGB images and LiDAR point clouds. Within a sequence, LiDAR scans are annotated every 5–10 s of driving. The sequences are recorded over the course of a year, resulting in a diverse set of weather conditions and outdoor environments. The TAS500 data set (Metzger et al. 2021) shows a higher percentage of vegetation when compared to the Freiburg Forest data set. Additionally, it includes data collected in many weather conditions, including both sunny and rainy settings, which enhances its generalizability. Table 6 summarizes all the available off-road datasets currently used for semantic terrain segmentation tasks.

Szabó and Weltsch (2024) have investigated the interoperability and cross-performance between two of the datasets. The study indicates that a CNN trained with the RUGD real data set can identify basic classes from the RELLIS-3D data set, such as tree and bush classes. The evaluation concluded that it is not the complexity of the classes that primarily impacts the training process, but rather the amount of data available for training. For both the RELLIS-3D and RUGD datasets, the neural network performs well on classes like grass, tree, sky, and mud, where there are sufficient examples in the training data set. However, it struggles with classes that have fewer instances in the training data. Researchers have also explored combining multiple datasets to achieve better generalization and performance across diverse environmental settings, yielding promising results (Walch et al. 2022).

5.2 | Synthetic Off-Road Datasets

Synthetic datasets have gained immense popularity in the past decade, and the synthetic data generation market is expected to reach \$2.1 billion by 2028 (Synthetic Data Generation Market to Skyrocket to 2.1 Billion by 2028—bccresearch.com 2025). This is due to advancements in computer graphics technology, which facilitate the generation of highly realistic environments. Modern rendering engines like Unity (Unity Real-Time Development Platform, 3D, 2D, VR & AR Engine—unity.com 2025) and Unreal Engine (The most powerful real-time 3D creation tool—unrealengine.com 2025) employ techniques, such as physically based rendering (PBR), global illumination, and high-dynamic-range (HDR) lighting to create visuals that closely resemble real-world scenes. These engines also simulate realistic physics interactions, including vehicle dynamics and dynamic weather effects, such as rain and snow. Integration of

AI enables intelligent behaviors, while procedural generation tools automate diverse environment creation. Tools like Blender (Blender Foundation 2025) further enhance modeling and animation capabilities, supporting collaborative development of synthetic datasets. These advancements mitigate cost and logistical challenges, offering precise experimentation for developing robust machine learning models in off-road terrain classification and perception tasks. One of the standout examples of leveraging synthetic environments for data set generation is the use of commercial video games like Grand Theft Auto V (GTAV), Watch Dogs, Red Dead Redemption 2, and so on. The realism of these computer-generated environments goes beyond just looking realistic. It is about how everything within the game world is meticulously designed and crafted: from the arrangement of objects and environments to the lifelike textures that cover them. The movement of vehicles and characters, including autonomous ones, adds to this authenticity.

Among game-generated datasets, GTASynth (Curnis et al. 2022) is unique because it is the only one dedicated to off-road/unstructured environment perception. This study uses a simulated vehicle driving across the vast map of Grand Theft Auto V to collect 3D data, such as point clouds and RGB images, using the DeepGTAV-PreSIL framework (Hurl et al. 2019). GTASynth includes a variety of outdoor natural environments, including mountains, forests, and fields; 16,207 pairs of artificial point clouds and images were captured while navigating through the game environment. RGB image data is generated by SynPhoRest (Nunes et al. 2022) and OffRoadSynth (Małek et al. 2024) using Unity Game Engine and the Perception Camera component (The Perception Camera Component, Perception Package, 1.0.0-preview.1—1.0 2025). SORT (Yan et al. 2022) gathers segmented image data and masks from desert, mountain, grassland, and forest areas using Unreal Engine 4 and an Unmanned Motorcycle Perspective.

Synthetic data, while increasingly valuable as a complement to real-world data, faces significant challenges due to domain discrepancies. For instance, Szabó and Weltsch (2024) demonstrated that a CNN trained solely on synthetic datasets fails to generalize effectively when confronted with inputs from real datasets, even for basic classes. This disparity arises from visual and dynamic differences between simulated and real-world environments.

The research area of domain adaptation (James et al. 2019; Bousmalis et al. 2018) and domain randomization (Tobin et al. 2017) requires further exploration to effectively address these challenges and enhance the applicability of synthetic data in real-world settings. Table 7 summarizes synthetic off-road datasets currently used for semantic terrain segmentation tasks.

6 | Off-Road Use Cases and Applications

Numerous applications of autonomous systems rely heavily on terrain classification as a fundamental tool for perception and safe navigation. As such, studies in the literature have shed light on several sectors where terrain classification plays a pivotal role. Terrain classification finds utility in various domains, each leveraging its capabilities for specific purposes.

TABLE 6 | Overview of real-world off-road datasets.

Reference	Data set type	Robot platform	Number of classes	Benchmark results (mIoU)	Data set description
Freiburg Forest (Valada et al. 2017)	RGB, multi-spectral, stereo	VIONA	6	SSMA- 84.18 AdapNet++ - 83.09 (Valada et al. 2020)	15,000 raw images Multi-spectrum/modality benchmark: 325 annotated RGB, Depth, NIR, NRG, NDVI, EVI variants.
RELLIS-3D (Jiang et al. 2021)	RGB, stereo, LiDAR	Warthog	20	GA-Nav - 74.44 (Guan et al. 2022) gscnn - 50.13 (Jiang et al. 2021)	13,556 LiDAR scans and 6,235 images.
RUGD (Wigness et al. 2019)	RGB	Husky	24	GA-Nav - 89.08 (Guan et al. 2022)	7436 images with 50032 labeled objects
OFFSED (Neigel et al. 2020)	RGB, stereo	ZED	19	—	203 annotated stereo images
YCOR (Maturana et al. 2017)	RGB	ATV	8	dark-fcn - 48.54 cnns-fcn - 46.66 (Maturana et al. 2017)	1076 images
TAS500 (Metzger et al. 2021)	RGB	MuCAR-3	24	ERFNet - 59.42 SQNet - 57.64 Fast-SCNN - 53 (Metzger et al. 2021)	640 annotated images
GOOSE (Mortimer et al. 2023)	RGB LiDAR	MuCAR-3	64	DDRNet - 46.53 PP-LiteSeg - 45.09 (Mortimer et al. 2023)	10,000 annotated pairs of RGB images and point clouds
CaT (Sharma et al. 2022)	RGB	ATV	3	PSPNet - ResNet101 - 80.57 PSPNet - ResNet34 - 80.12 (Sharma et al. 2022)	1812 annotated images
BotanicGarden (Liu et al. 2024c)	RGB, Stereo	Scout V1.0	27	—	1181 annotated images

TABLE 7 | Overview of synthetic off-road datasets.

Reference	Data collection method	Data set type	Data set description
GTASynth (Curnis et al. 2022)	DeepGTAV	Camera and LiDAR	16.207 point clouds and images
SynPhoRest (Nunes et al. 2022)	Unity game engine	Camera and LiDAR	3154 frames for six semantic classes
SORT (Yan et al. 2022)	Unreal Engine 4	Camera	20,844 scenes
EDEN (Le et al. 2021)	Blender	Camera	30,000 images
OffRoadSynth (Malek et al. 2024)	Unity game engine	Camera	6170 RGB images and ground truth labels

6.1 | Planetary Exploration

Planetary exploration rovers, deployed in extraterrestrial environments such as Mars, heavily rely on terrain classification for safe navigation and scientific exploration. Across various celestial bodies, terrain classification plays a critical role in identifying and navigating diverse terrain types, such as dunes, canyons, craters, volcanoes, and rocky surfaces. Accurate terrain classification allows rovers to make informed decisions about their movements, adjust their trajectories according to the terrain's characteristics, and avoid potential hazards like steep slopes or unstable surfaces. This capability is crucial not only for ensuring the safety and efficiency of robotic missions but also for maximizing the scientific returns from these explorations. By categorizing and mapping planetary terrains, rovers can contribute to our understanding of planetary geology, climate history, and potential resources. Terrain classification is pivotal for future habitability research across different planetary surfaces.

By identifying areas with favorable terrain characteristics, such as stable surfaces or accessible slopes, scientists can prioritize these regions for future human exploration missions and potential habitat locations. Lots of terrain classification studies are found in the literature, where methods and algorithms are fine-tuned, optimized, and used for planetary rovers (Lv et al. 2022; Ono et al. 2015; Brooks and Iagnemma 2012; Hewitt et al. 2017; Halatci et al. 2007). Key datasets like MSLNet (Mars surface image (Curiosity rover) labeled data set—zenodo.org 2025) and AI4MARS (Swan et al. 2021) are instrumental in advancing Martian terrain classification and perception. MSLNet comprises 6691 images collected by the Mars Science Laboratory (MSL) rover, Curiosity, manufactured by NASA. AI4MARS includes approximately 326,000 semantic segmentation labels covering 35,000 images from rovers, such as Curiosity, Opportunity, and Spirit, obtained through crowd-sourcing. Continued research and development in terrain classification methods will drive further advancements in planetary exploration robotics, facilitating future missions to explore new frontiers and unlock the mysteries of the universe.

6.2 | Military Applications

In military contexts, terrain classification aids in route planning, reconnaissance, surveillance, and target detection. By accurately classifying different types of terrain, such as forests, urban areas, deserts, and mountainous regions, military vehicles can navigate safely, avoid hazards, and maintain stealth when necessary.

Additionally, terrain classification enhances situational awareness for military personnel, providing valuable insights into enemy movements and potential threats. Studies in the literature have explored terrain classification methods tailored specifically for military applications (Pokonieczny 2018; Vecherin et al. 2020; Ahmadi et al. 2022). These studies often focus on optimizing algorithms for real-time processing, integrating data from multiple sensors, and enhancing classification accuracy in dynamic and unpredictable environments. By leveraging advanced technologies, such as machine learning, computer vision, and sensor fusion, military-grade terrain classification systems can operate autonomously and adaptively in diverse operational scenarios. Continued research in this area will further enhance the capabilities of military platforms and contribute to the development of more robust and intelligent defense systems.

6.3 | Agriculture

Recently, agriculture has seen an increased use of automated machinery for various tasks, such as seeding, planting, harvesting, and crop monitoring (Rahmadian and Widyartono 2020). Successful research in the development of autonomous agricultural robots has been demonstrated in various studies, including those on citrus picking (Aloisio et al. 2012), strawberry harvesting (Qingchun et al. 2012), and tomato harvesting robots (Jun et al. 2012). However, effective deployment of these robots heavily depends on robust terrain perception, as agricultural environments consist of diverse and often unstructured terrains, including loose soil, muddy fields, dense vegetation, and sloped surfaces. Accurate terrain classification enables robots to assess traversability, optimize navigation paths, and prevent operational failures caused by terrain-induced slippage or immobilization. By leveraging real-time sensor data, these robots can make informed decisions to navigate challenging landscapes, ensuring smooth operation and improved efficiency in agricultural tasks.

7 | Future Trends and Gaps in Off-Road Terrain Classification

7.1 | Data Set Diversity, Annotation, and Labeling

A primary challenge in off-road terrain classification is the scarcity of diverse and accurately labeled datasets tailored to various conditions. Existing datasets, such as the Freiburg Forest, RUGD, and RELLIS-3D, offer valuable resources but are limited in scope. Each environment poses unique challenges,

complicating the generalization of models across different terrains. Manual annotation is challenging due to the complexity and variability of off-road environments, which often include diverse terrain types and conditions. This variability leads to inconsistencies and errors in annotations, impacting the quality and reliability of training datasets. The CityScape data set required an average annotation time of more than 90 min per frame (Cordts et al. 2016). To address these challenges, researchers are increasingly turning to AI-driven approaches for automatic and semi-automatic annotation (Sumathi et al. 2011; Adnan et al. 2021). AI algorithms, powered by deep learning techniques, can analyze large volumes of data and annotate images more efficiently than manual methods. Unsupervised and weakly supervised learning methods present alternative approaches to reduce reliance on manually labeled data. These techniques leverage unlabeled or weakly labeled data to train models, relying on intrinsic data structures or minimal supervision to infer meaningful patterns and labels. Future research should focus on advancing AI-driven annotation techniques and exploring novel learning paradigms to enhance data set quality and diversity.

7.2 | Computational Complexity

The integration of multiple sensor modalities and advanced algorithms in hybrid terrain classification methods significantly increases computational complexity. This heightened complexity poses challenges for real-time deployment, especially in resource-constrained environments or embedded systems. The corresponding costs of machine learning-based training also escalate due to increased computational demands.

An effective solution involves co-optimizing the internal structures of convolutional layers (Maji and Mullins 2018) and the introduction of optimization models by fine-tuning algorithmic parameters and architecture (Im and Dasari 2022). Techniques such as pruning and quantization also hold potential for optimizing computational efficiency in terrain classification. While these methods are valuable, their detailed exploration falls beyond the scope of this paper (refer to (Hossain et al. 2023) for a better understanding). Further research is needed to refine existing techniques and explore novel approaches that balance computational demands with the performance requirements of autonomous systems.

7.3 | Adaptability and Generalization

Off-road terrain classification models must demonstrate adaptability and generalization across diverse and dynamic environments. Variations in terrain types, lighting conditions, weather patterns, and seasonal changes present significant challenges for model robustness. Domain discrepancy remains a critical issue, as highlighted in Section 5.2, where models trained on one data set struggle to perform effectively on another type of input. Domain adaptation techniques offer potential solutions (Truong et al. 2021). Domain alignment aims to reduce the gap between synthetic and real data distributions through techniques like adversarial training. Domain

randomization introduces variability during synthetic data generation to better simulate real-world conditions (Tobin et al. 2017).

Moving forward, future research should focus on addressing these challenges and advancing methodologies for off-road terrain classification. Improving data set diversity and quality through enhanced annotation techniques and automated labeling processes will be crucial. Additionally, further exploration of domain adaptation techniques is essential to bridge the gap between synthetic and real-world data. Novel approaches integrating multi-modal sensor data and advanced machine learning algorithms hold promise for enhancing the robustness and applicability of terrain classification models in diverse and challenging environments.

8 | Conclusion

In this review, we delve into the advancements, challenges, and future trajectories of terrain classification methods for off-road perception. The research landscape in terrain classification has evolved significantly, driven by technological advancements, algorithmic innovations, and the growing demand for autonomous systems capable of navigating unstructured environments. Our examination of the literature highlights a shift toward self-supervised learning as a promising solution to mitigate data scarcity challenges in off-road terrain classification. While synthetic datasets have been widely utilized, they are not yet sufficiently reliable for real-world deployment. However, synthetic data generation is an emerging and promising domain for expanding the available data pool. With advancements in hyper-realistic simulation environments, improved computational hardware, and domain adaptation techniques, the gap between synthetic and real-world data is gradually narrowing. Future research should focus on developing deep learning methods that can effectively learn from unlabeled data while remaining computationally efficient. Leveraging high-fidelity synthetic data alongside self-supervised techniques could significantly enhance the adaptability and robustness of terrain classification models for practical applications. By addressing the challenges and pursuing interdisciplinary research endeavors, we can develop robust, scalable, and real-time solutions for autonomous navigation, environmental monitoring, and exploration in challenging off-road environments.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The authors have nothing to report.

References

- Adnan, M. M., M. S. M. Rahim, A. Rehman, Z. Mehmood, T. Saba, and R. A. Naqvi. 2021. "Automatic Image Annotation Based on Deep Learning Models: A Systematic Review and Future Challenges." *IEEE Access* 9: 50253–50264. <https://doi.org/10.1109/ACCESS.2021.3068897>.
- Agishev, R., T. Petříček, and K. Zimmermann. 2022. "Trajectory Optimization Using Learned Robot-Terrain Interaction Model in Exploration of Large Subterranean Environments." *IEEE Robotics and Automation Letters* 7, no. 2: 3365–3371. <https://doi.org/10.1109/LRA.2022.3147332>.
- Ahmadi, K. D., A. J. Rashidi, and A. M. Moghri. 2022. "Design and Simulation of Autonomous Military Vehicle Control System Based on Machine Vision and Ensemble Movement Approach." *Journal of Supercomputing* 78, no. 15: 17309–17347.
- Ahtiainen, J., T. Stoyanov, and J. Saarinen. 2017. "Normal Distributions Transform Traversability Maps: Lidar-Only Approach for Traversability Mapping in Outdoor Environments." *Journal of Field Robotics* 34, no. 3: 600–621.
- Aloisio, C., R. K. Mishra, C.-Y. Chang, and J. English. 2012. "Next Generation Image Guided Citrus Fruit Picker." In 2012 IEEE International Conference On Technologies for Practical Robot Applications (TePRA), 37–41.
- Anand, B., H. Verma, A. Thakur, P. Alam, and P. Rajalakshmi. 2022. "Evaluation of The Quality Of Lidar Data In The Varying Ambient Light." In 2022 IEEE Sensors Applications Symposium (SAS), 1–5. <https://doi.org/10.1109/SAS54819.2022.9881373>.
- Androulakis, V., J. Sottile, S. Schafrik, and Z. Agioutantis. 2020. "Concepts for Development of Autonomous Coal Mine Shuttle Cars." *IEEE Transactions on Industry Applications* 56, no. 3: 3272–3280. <https://doi.org/10.1109/TIA.2020.2972786>.
- Anguelov, D., B. Taskarf, V. Chatalbashev, et al. 2005. "Discriminative Learning of Markov Random Fields for Segmentation of 3d Scan Data." In 2005 IEEE Computer Society Conference on Computer Vision and Pattern Recognition (CVPR'05) Vol. 2, 169–176.
- Badrinarayanan, V., A. Kendall, and R. Cipolla. 2017. "Segnet: A Deep Convolutional Encoder-Decoder Architecture for Image Segmentation." *IEEE Transactions on Pattern Analysis and Machine Intelligence* 39, no. 12: 2481–2495.
- Bai, C., J. Guo, L. Guo, and J. Song. 2019. "Deep Multi-Layer Perception Based Terrain Classification for Planetary Exploration Rovers." *Sensors* 19, no. 14: 3102. <https://doi.org/10.3390/s19143102>.
- Bajracharya, M., A. Howard, L. H. Matthies, B. Tang, and M. Turmon. 2009. "Autonomous Off-Road Navigation With End-to-End Learning for the Lagr Program." *Journal of Field Robotics* 26, no. 1: 3–25.
- Bay, H., T. Tuytelaars, and L. van Gool. 2006. "SURF: Speeded Up Robust Features." *Computer Vision – ECCV 2006*: 404–417. https://doi.org/10.1007/11744023_32.
- Bellone, M., G. Reina, L. Caltagirone, and M. Wahde. 2018. "Learning Traversability From Point Clouds in Challenging Scenarios." *IEEE Transactions on Intelligent Transportation Systems* 19, no. 1: 296–305. <https://doi.org/10.1109/TITS.2017.2769218>.
- Beycimén, S., A. Art, M. M. Unal, D. Ignatyev, and A. Zolotas. 2024. "On Real-Time Semantic Segmentation With Comprehensive Off-Road Datasets for Enhanced Terrain Classification." Available at SSRN 4947521.
- Beycimén, S., D. Ignatyev, and A. Zolotas. 2023. "A Comprehensive Survey of Unmanned Ground Vehicle Terrain Traversability for Unstructured Environments and Sensor Technology Insights." *Engineering Science and Technology, an International Journal* 47: 101457. <https://doi.org/10.1016/j.jestech.2023.101457>.
- Blender Foundation. 2025. blender.org—Home of the Blender Project—Free and Open 3D Creation Software—blender.org. <https://www.blender.org/>.
- Borges, P. V., T. Peynot, S. Liang, et al. 2022. "A Survey on Terrain Traversability Analysis for Autonomous Ground Vehicles: Methods, Sensors, and Challenges." *Field Robotics* 2, no. 1: 1567–1627.
- Bousmalis, K., A. Irpan, P. Wohlhart, et al. 2018. "Using Simulation and Domain Adaptation to Improve Efficiency of Deep Robotic Grasping." In 2018 IEEE International Conference on Robotics And Automation (ICRA), 4243–4250.
- Bradley, D. M., J. K. Chang, D. Silver, et al. 2015. "Scene Understanding for a High-Mobility Walking Robot." In 2015 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 1144–1151.
- Broggi, A., E. Cardarelli, S. Cattani, and M. Sabbatelli. 2013. "Terrain Mapping for Off-Road Autonomous Ground Vehicles Using Rational B-Spline Surfaces and Stereo Vision." In 2013 IEEE Intelligent Vehicles Symposium (IV), 648–653.
- Brooks, C. A., and K. Iagnemma. 2005. "Vibration-Based Terrain Classification for Planetary Exploration Rovers." *IEEE Transactions on Robotics* 21, no. 6: 1185–1191.
- Brooks, C. A., and K. Iagnemma. 2012. "Self-Supervised Terrain Classification for Planetary Surface Exploration Rovers." *Journal of Field Robotics* 29, no. 3: 445–468.
- Cai, H., J. Li, M. Hu, C. Gan, and S. Han. 2023. "Efficientvit: Lightweight Multi-Scale Attention For High-Resolution Dense Prediction." In Proceedings of the IEEE/CVF International Conference On Computer Vision, 17302–17313.
- Caron, M., H. Touvron, I. Misra, et al. 2021. "Emerging Properties in Self-Supervised Vision Transformers." In Proceedings of the IEEE/CVF International Conference On Computer Vision, 9650–9660.
- Castro, M. G., S. Triest, W. Wang, et al. 2023. "How Does It Feel? Self-Supervised Costmap Learning For Off-Road Vehicle Traversability." In 2023 IEEE International Conference on Robotics and Automation (ICRA), 931–938.
- Chambers, D. R., J. Gassaway, C. Goodin, and P. J. Durst. 2015. "Simulation of a Multispectral, Multicamera, Off-Road Autonomous Vehicle Perception System With Virtual Autonomous Navigation Environment (Vane)." In Electro-Optical and Infrared Systems: Technology and Applications XII; and Quantum Information Science and Technology Vol. 9648, 964802.
- Chatzichristofis, S. A., and Y. S. Boutalis. 2008. "Cedd: Color and Edge Directivity Descriptor: A Compact Descriptor For Image Indexing And Retrieval." In Computer Vision Systems: 6th International Conference, ICVS 2008 Santorini, Greece, May 12–15, 2008 Proceedings 6, 312–322.
- Chemhengcharoen, P., P. Nilsumrit, P. Pongpetrarat, and G. Phanomchoeng. 2019. "Development of a Prototype of Autonomous Vehicle For Agriculture Applications." In Proceedings of the 7th International Conference on Communications and Broadband Networking, 53–57. Association for Computing Machinery. <https://doi.org/10.1145/3330180.3330191>.
- Chen, E., C. Ho, M. Maulimov, C. Wang, and S. Scherer. 2024. "Learning-on-the-Drive: Self-Supervised Adaptive Long-Range Perception for High-Speed Offroad Driving." In 2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 9588–9595.
- Chen, H., and X. Lan. 2024. "Real-Time Semantic Segmentation Based on bisenetv2 for Wild Road." *Journal of Intelligent Systems* 33, no. 1: 20230205.
- Chen, L.-C., G. Papandreou, I. Kokkinos, K. Murphy, and A. L. Yuille. 2017. "DeepLab: Semantic Image Segmentation With Deep Convolutional Nets, Atrous Convolution, and Fully Connected CRFS." *IEEE Transactions on Pattern Analysis and Machine Intelligence* 40, no. 4: 834–848.
- Chetan, J., M. Krishna, and C. Jawahar. 2010a. "Fast and Spatially-Smooth Terrain Classification Using Monocular Camera." In 2010 20th International Conference on Pattern Recognition, 4060–4063.

- Chetan, J., K. Madhava Krishna, and C. V. Jawahar. 2010b. "An Adaptive Outdoor Terrain Classification Methodology Using Monocular Camera." In 2010 IEEE/RSJ International Conference on Intelligent Robots and Systems, 766–771. <https://doi.org/10.1109/IROS.2010.5651067>.
- Clark, A., J. Simpson, and J. Hall. 2019. "Comparing CNN Inputs for Terrain Classification Using Simulation." In 2019 First International Conference on Transdisciplinary AI (TransAI), 43–47. <https://doi.org/10.1109/TransAI46475.2019.00015>.
- Condé Nast. 2025. Self-Driving Cars Are Being Put on a Data Diet—wired.com. <https://www.wired.com/story/self-driving-cars-are-being-put-on-a-data-diet/>.
- Connected Papers. 2025. Connected Papers: A Unique Graph-Based Tool to Explore and Discover Papers Relevant to Your Field of Work. <https://www.connectedpapers.com/>.
- Cordts, M., M. Omran, S. Ramos, et al. 2016. "The Cityscapes Dataset for Semantic Urban Scene Understanding." In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 3213–3223.
- Cortinhal, T., G. Tzelepis, and E. Erdal Aksoy. 2020. "Salsanext: Fast, Uncertainty-Aware Semantic Segmentation of Lidar Point Clouds." In Advances in Visual Computing: 15th International Symposium, ISVC 2020, San Diego, CA, USA, October 5–7, 2020, Proceedings, Part II 15, 207–222.
- Csik, D., A. Odry, J. Sarosi, and P. Sarcevic. 2021. "Inertial Sensor-Based Outdoor Terrain Classification For Wheeled Mobile Robots." In 2021 IEEE 19th International Symposium on Intelligent Systems and Informatics (SISY), 159–164. <https://doi.org/10.1109/SISY52375.2021.9582504>.
- Curnis, G., S. Fontana, and D. G. Sorrenti. 2022. "Gtasynt: 3d Synthetic Data of Outdoor Non-Urban Environments." *Data Brief* 43: 108412.
- CVAT—cvat.ai. 2025. <https://www.cvat.ai/>.
- Dabbiru, L., C. Goodin, N. Scherrer, and D. Carruth. 2020. "Lidar Data Segmentation in Off-Road Environment Using Convolutional Neural Networks (CNN)." *SAE International Journal of Advances and Current Practices in Mobility* 2: 3288–3292.
- Dabbiru, L., S. Sharma, C. Goodin, et al. 2021. "Traversability Mapping in Off-Road Environment Using Semantic Segmentation." In Autonomous Systems: Sensors, Processing, and Security for Vehicles and Infrastructure 2021 Vol. 11748: 78–83.
- Dahlkamp, H., A. Kaehler, D. Stavens, S. Thrun, and G. R. Bradski. 2006. "Self-Supervised Monocular Road Detection In Desert Terrain." In Robotics: Science and Systems Vol. 38.
- Dai, X., S. Li, and F. Sun. 2014. "Adaptive Terrain Classification in Field Environment Based On Self-Supervised Learning." In Proceedings of 2014 IEEE Chinese Guidance, Navigation and Control Conference, 6–11. <https://doi.org/10.1109/CGNCC.2014.7007211>.
- Dargazany, A., and K. Berns. 2014. "Stereo-Based Terrain Traversability Estimation Using Surface Normals." In ISR/Robotik 2014; 41st International Symposium on Robotics, 1–7.
- Dargazany, A. R. 2019. "Stereo-Based Terrain Traversability Analysis Using Normal-Based Segmentation And Superpixel Surface Analysis." arXiv preprint arXiv:1907.06823.
- DARPA Grand Challenge. 2005. DARPA Grand Challenge—Wikipedia. https://en.wikipedia.org/wiki/DARPA_Grand_Challenge.
- Deng, J., W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei. 2009. "Imagenet: A Large-Scale Hierarchical Image Database." In 2009 IEEE Conference On Computer Vision And Pattern Recognition, 248–255.
- Deoli, P., S. A. Deshpande, A. Vierling, and K. Berns. 2024a. "Exploring Image Fusion Techniques For Off-Road Semantic Segmentation In Harsh Lighting Conditions. A Multispectral Imagery Analysis." In 2024 21st International Conference on Ubiquitous Robots (UR), 566–573.
- Deoli, P., R. Kumar, A. Vierling, and K. Berns. 2024b. "Evaluating the Robustness of Off-Road Autonomous Driving Segmentation Against Adversarial Attacks: A Dataset-Centric Analysis." arXiv preprint arXiv:2402.02154.
- Dinh, L., J. Sohl-Dickstein, and S. Bengio. 2016. "Density Estimation Using Real NVP." arXiv preprint arXiv:1605.08803.
- Dosovitskiy, A., L. Beyer, A. Kolesnikov, et al. 2020. "An Image Is Worth 16x16 Words: Transformers For Image Recognition at Scale." arXiv preprint arXiv:2010.11929.
- Douillard, B., J. Underwood, N. Kuntz, et al. 2011. "On the Segmentation of 3d Lidar Point Clouds." In 2011 IEEE International Conference on Robotics and Automation, 2798–2805.
- Douillard, B., J. Underwood, N. Melkumyan, et al. 2010. "Hybrid Elevation Maps: 3d Surface Models for Segmentation." In 2010 IEEE/RSJ International Conference on Intelligent Robots and Systems, 1532–1538.
- Dunbabin, M., and L. Marques. 2012. "Robots for Environmental Monitoring: Significant Advancements and Applications." *IEEE Robotics & Automation Magazine* 19, no. 1: 24–39. <https://doi.org/10.1109/MRA.2011.2181683>.
- Dupont, E. M., C. A. Moore, E. G. Collins, and E. Coyle. 2008. "Frequency Response Method for Terrain Classification in Autonomous Ground Vehicles." *Auton Robots* 24: 337–347.
- Dutta, A., and P. Dasgupta. 2017. "Ensemble Learning With Weak Classifiers for Fast and Reliable Unknown Terrain Classification Using Mobile Robots." *IEEE Transactions on Systems, Man, and Cybernetics: Systems* 47, no. 11: 2933–2944. <https://doi.org/10.1109/TSMC.2016.2531700>.
- Dutta, A., and A. Zisserman. 2019. "The Via Annotation Software for Images, Audio and Video." In Proceedings Of The 27th ACM International Conference on Multimedia, 2276–2279.
- Elander, F. 2021. Semantic Segmentation Of Off-Road Scenery On Embedded Hardware Using Transfer Learning.
- Error-riddled data sets are warping our sense of how good AI really is—technologyreview.com. 2025. <https://www.technologyreview.com/2021/04/01/1021619/ai-data-errors-warp-machine-learning-progress/>.
- Faykus, III, M. H., A. Pickeral, E. Marquez, M. C. Smith, and J. C. Calhoun. 2024. "Efficient Vision Transformers for Autonomous Off-Road Perception Systems." *Journal of Computer and Communications* 12, no. 9: 188–207.
- Filitchkin, P., and K. Byl. 2012. "Feature-Based Terrain Classification For Litledog." In 2012 IEEE/RSJ International Conference on Intelligent Robots and Systems, 1387–1392. <https://doi.org/10.1109/IROS.2012.6386042>.
- Firkat, E., F. An, B. Peng, et al. 2023. "Fgseg: Field-Ground Segmentation for Agricultural Robot Based on Lidar." *Computers and Electronics in Agriculture* 211: 107965.
- Fischler, M. A., and R. C. Bolles. 1987. "Random Sample Consensus: A Paradigm For Model Fitting With Applications To Image Analysis And Automated Cartography." In Readings in Computer Vision, edited by M. A. Fischler and O. Firschein, 726–740. Morgan Kaufmann. <https://doi.org/10.1016/B978-0-08-051581-6.50070-2>.
- Forkel, B., J. Kallwies, and H.-J. Wuensche. 2021. "Probabilistic Terrain Estimation For Autonomous Off-Road Driving." In 2021 IEEE International Conference on Robotics and Automation (ICRA), 13864–13870.
- Fox, D., W. Burgard, and S. Thrun. 1997. "The Dynamic Window Approach to Collision Avoidance." *IEEE Robotics & Automation Magazine* 4, no. 1: 23–33.
- Frey, J., M. Mattamala, N. Chebrolu, C. Cadena, M. Fallon, and M. Hutter. 2023. "Fast Traversability Estimation For Wild Visual Navigation." arXiv preprint arXiv:2305.08510.

- Frey, J., M. Patel, D. Atha, et al. 2024. "Roadrunner—Learning Traversability Estimation for Autonomous Off-Road Driving." *IEEE Transactions on Field Robotics* 1: 192–212. <https://doi.org/10.1109/tfr.2024.3464369>.
- Fu, J., J. Liu, H. Tian, et al. 2019. "Dual Attention Network For Scene Segmentation." In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 3146–3154.
- Gao, B., A. Xu, Y. Pan, X. Zhao, W. Yao, and H. Zhao. 2019. "Off-Road Drivable Area Extraction Using 3d Lidar Data." In 2019 IEEE Intelligent Vehicles Symposium (IV), 1505–1511.
- Gasparino, M. V., A. N. Sivakumar, Y. Liu, et al. 2022. "Wayfast: Navigation With Predictive Traversability in the Field." *IEEE Robotics and Automation Letters* 7, no. 4: 10651–10658.
- Geiger, A., P. Lenz, C. Stiller, and R. Urtasun. 2013. "Vision Meets Robotics: The Kitti Dataset." *The International Journal of Robotics Research* 32: 1231–1237. <https://doi.org/10.1177/0278364913491297>.
- GitHub - googlecreativelab/quickdraw-dataset: Documentation on how to access and use the Quick, Draw! Dataset.—github.com. 2025. <https://github.com/googlecreativelab/quickdraw-dataset>.
- Guan, T., D. Kothandaraman, R. Chandra, A. J. Sathyamoorthy, K. Weerakoon, and D. Manocha. 2022. "Ga-Nav: Efficient Terrain Segmentation for Robot Navigation in Unstructured Outdoor Environments." *IEEE Robotics and Automation Letters* 7, no. 3: 8138–8145.
- Guastella, D. C., and G. Muscato. 2020. "Learning-Based Methods of Perception and Navigation for Ground Vehicles In Unstructured Environments: A Review." *Sensors* 21, no. 1: 73.
- Guo, C., W. Sato, L. Han, S. Mita, and D. McAllester. 2011. "Graph-Based 2d Road Representation of 3d Point Clouds For Intelligent Vehicles." In 2011 IEEE Intelligent Vehicles Symposium (IV), 715–721.
- Gutierrez, J., J. Gordillo, and I. Lopez. 2004. "Configuration and Construction of an Autonomous Vehicle For A Mining Task." In IEEE International Conference on Robotics and Automation, 2004. Proceedings. ICRA '04. 2004 Vol. 2: 2010–2016. <https://doi.org/10.1109/ROBOT.2004.1308119>.
- Halatci, I., C. A. Brooks, and K. Iagnemma. 2007. "Terrain Classification and Classifier Fusion For Planetary Exploration Rovers." In 2007 IEEE Aerospace Conference, 1–11.
- Hanson, J. 2011. "Land Rover's Terrain Response Explained – Exploring Overland." Retrieved from <https://www.exploringoverland.com/overland-tech-travel/2011/7/28/land-rovers-terrain-response-explained.html>.
- Hassan, B., N. A. Madjid, F. Kashwani, M. Alansari, M. Khonji, and J. Dias. 2024a. "Pathformer: A Transformer-Based Framework For Vision-Centric Autonomous Navigation in Off-Road Environments." In 2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 7718–7725.
- Hassan, B., A. Sharma, N. A. Madjid, M. Khonji, and J. Dias. 2024b. "Terrainsense: Vision-Driven Mapless Navigation For Unstructured Off-Road Environments." In 2024 IEEE International Conference on Robotics and Automation (ICRA), 18229–18235.
- Hassan, M. U., M. Ullah, and J. Iqbal. 2016. "Towards Autonomy in Agriculture: Design and Prototyping Of A Robotic Vehicle With Seed Selector." In 2016 2nd International Conference on Robotics and Artificial Intelligence (ICRAI), 37–44. <https://doi.org/10.1109/ICRAI.2016.7791225>.
- He, K., X. Zhang, S. Ren, and J. Sun. 2016. "Deep Residual Learning For Image Recognition." In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 770–778.
- Hewitt, R. A., A. Ellery, and A. de Ruiter. 2017. "Training a Terrain Traversability Classifier for a Planetary Rover Through Simulation." *International Journal of Advanced Robotic Systems* 14, no. 5: 1729881417735401.
- Himmelsbach, M., F. V. Hundelshausen, and H.-J. Wuensche. 2010. "Fast Segmentation Of 3d Point Clouds for Ground Vehicles." In 2010 IEEE Intelligent Vehicles Symposium, 560–565.
- Holder, C. J., T. P. Breckon, and X. Wei. 2016. "From On-Road to Off: Transfer Learning Within a Deep Convolutional Neural Network for Segmentation and Classification of Off-Road Scenes." In Computer Vision—ECCV 2016 Workshops: Amsterdam, The Netherlands, October 8–10 and 15–16, 2016, Proceedings, Part I 14, 149–162.
- Hossain, M. B., N. Gong, and M. Shaban. 2023. "Computational Complexity Reduction Techniques for Deep Neural Networks: A Survey." In 2023 IEEE International Conference on Artificial Intelligence, Blockchain, and Internet of Things (AIBThings), 1–6. <https://doi.org/10.1109/AIBThings58340.2023.10292477>.
- Howard, A., M. Turmon, L. Matthies, B. Tang, A. Angelova, and E. Mjolsness. 2006. "Towards Learned Traversability for Robot Navigation: From Underfoot to the Far Field." *Journal of Field Robotics* 23, no. 11–12: 1005–1017.
- Hu, Q., B. Yang, L. Xie, et al. 2020. "Randla-net: Efficient Semantic Segmentation Of Large-Scale Point Clouds." In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 11108–11117.
- Huang, X., X. Cheng, Q. Geng, et al. 2018. "The Apolloscape Dataset For Autonomous Driving." In 2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW), 1067–10676. <https://doi.org/10.1109/CVPRW.2018.00141>.
- Humblot-Renaux, G., L. Marchegiani, T. B. Moeslund, and R. Gade. 2022. "Navigation-Oriented Scene Understanding for Robotic Autonomy: Learning to Segment Driveability in Egocentric Images." *IEEE Robotics and Automation Letters* 7, no. 2: 2913–2920.
- Hurl, B., K. Czarnecki, and S. Waslander. 2019. "Precise Synthetic Image And Lidar (Presil) Dataset For Autonomous Vehicle Perception." In 2019 IEEE Intelligent Vehicles Symposium (IV), 2522–2529.
- Ibanez-Guzman, J., and A. Malcolm. 2002. "Autonomous Vehicles in the Construction Process." *Construction Innovation: Information, Process, Management* 2: 211–224. <https://doi.org/10.1108/14714170210814775>.
- Im, M. S., and V. R. Dasari. 2022. "Computational Complexity Reduction Of Deep Neural Networks." arXiv preprint arXiv:2207.14620.
- Intel Editorial: For Self-Driving Cars, There's Big Meaning Behind One Big Number: 4 Terabytes—businesswire.com. 2025. <https://www.businesswire.com/news/home/20170414005225/en/Intel-Editorial-For-Self-Driving-Cars-There%E2%80%99s-Big-Meaning-Behind-One-Big-Number-4-Terabytes#:~:text=Four%20terabytes%20is%20the%20estimated,in%20their%20car%20each%20day>.
- Islam, S., M. S. Tanvir, M. R. Habib, et al. 2022. "Autonomous Driving Vehicle System Using Lidar Sensor." In Intelligent Data Communication Technologies and Internet of Things, edited by D. J. Hemanth, D. Pelusi, C. Vuppapapati, 345–358, Springer Nature Singapore.
- Jackel, L. D., E. Krotkov, M. Perschbacher, J. Pippine, and C. Sullivan. 2006. "The Darpa Lagr Program: Goals, Challenges, Methodology, and Phase I Results." *Journal of Field Robotics* 23, no. 11–12: 945–973.
- James, S., P. Wohlhart, M. Kalakrishnan, et al. 2019. "Sim-To-Real Via Sim-to-Sim: Data-Efficient Robotic Grasping via Randomized-To-Canonical Adaptation Networks." In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 12627–12637.
- Jeon, Y., E. I. Son, and S.-W. Seo. 2024. "Follow the Footprints: Self-Supervised Traversability Estimation for Off-Road Vehicle Navigation Based on Geometric and Visual Cues." In 2024 IEEE International Conference on Robotics and Automation (ICRA), 1774–1780.
- Jiang, P., P. Osteen, M. Wigness, and S. Saripalli. 2021. "Relis-3D Dataset: Data, Benchmarks And Analysis." In 2021 IEEE International Conference On Robotics And Automation (ICRA), 1110–1116.

- Jin, Y., D. Han, and H. Ko. 2021. "Memory-Based Semantic Segmentation For Off-Road Unstructured Natural Environments." In 2021 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 24–31.
- Joho, D., C. Stachniss, P. Pfaff, and W. Burgard. 2007. "Autonomous Exploration For 3d Map Learning." In *Autonome Mobile Systeme 2007*: 20. Fachgespräch Kaiserslautern, 18./19. Oktober 2007, 22–28.
- Jun, W., Z. Zhou, and D. Xiaodong. 2012. "Design and Co-Simulation for Tomato Harvesting Robots." In *Proceedings of the 31st Chinese Control Conference*, 5105–5108.
- Jung, S., J. Lee, X. Meng, B. Boots, and A. Lambert. 2024. "V-Strong: Visual Self-Supervised Traversability Learning for Off-Road Navigation." In 2024 IEEE International Conference on Robotics and Automation (ICRA), 1766–1773.
- Kahn, G., P. Abbeel, and S. Levine. 2021. "Badgr: An Autonomous Self-Supervised Learning-Based Navigation System." *IEEE Robotics and Automation Letters* 6, no. 2: 1312–1319.
- Karnan, H., E. Yang, D. Farkash, G. Warnell, J. Biswas, and P. Stone. 2023. "Sterling: Self-Supervised Terrain Representation Learning From Unconstrained Robot Experience." arXiv preprint arXiv:2309.15302.
- Keep or Not to Keep...Data for Self-Driving Vehicles? — drivesweden.net. 2025. <https://www.drivesweden.net/en/news/keep-or-not-keepdata-self-driving-vehicles>. [Accessed 12-07-2024].
- Khan, Y. N., P. Komma, K. Bohlmann, and A. Zell. 2011a. "Grid-Based Visual Terrain Classification For Outdoor Robots Using Local Features." In 2011 IEEE Symposium On Computational Intelligence In Vehicles And Transportation Systems (CIVTS) Proceedings, 16–22.
- Khan, Y. N., P. Komma, and A. Zell. 2011b. "High Resolution Visual Terrain Classification For Outdoor Robots." In 2011 IEEE International Conference on Computer Vision Workshops (ICCV Workshops), 1014–1021.
- Kim, D., S. M. Oh, and J. M. Rehg. 2007. "Traversability Classification For UGV Navigation: A Comparison Of Patch And Superpixel Representations." In 2007 IEEE/RSJ International Conference on Intelligent Robots and Systems, 3166–3173. <https://doi.org/10.1109/IROS.2007.4399610>.
- Kim, J., D. Kim, J. Lee, J. Lee, H. Joo, and I.-S. Kweon. 2009. "Non-Contact Terrain Classification For Autonomous Mobile Robot." In 2009 IEEE International Conference on Robotics and Biomimetics (ROBIO), 824–829. <https://doi.org/10.1109/ROBIO.2009.5420568>.
- Kirillov, A., E. Mintun, N. Ravi, et al. 2023. "Segment Anything." In *Proceedings Of The IEEE/CVF International Conference On Computer Vision*, 4015–4026.
- Koenig, N., and A. Howard. 2004. "Design And Use Paradigms For Gazebo, An Open-Source Multi-Robot Simulator." In 2004 IEEE/RSJ International Conference On Intelligent Robots And Systems (IROS) (IEEE Cat. No. 04CH37566) Vol. 3: 2149–2154.
- Kragh, M., R. N. Jørgensen, and H. Pedersen. 2015. "Object Detection and Terrain Classification in Agricultural Fields Using 3d Lidar Data." In *International Conference On Computer Vision Systems*, 188–197.
- Krebs, A., C. Pradalier, and R. Siegwart. 2009. "Comparison of Boosting Based Terrain Classification Using Proprioceptive And Exteroceptive Data." In *Experimental Robotics: The Eleventh International Symposium*, 93–102.
- Kurobe, A., Y. Nakajima, K. Kitani, and H. Saito. 2021. "Audio-Visual Self-Supervised Terrain Type Recognition for Ground Mobile Platforms." *IEEE Access* 9: 29970–29979.
- Kurup, A., S. Kysar, and J. Bos. 2020. "SVM-Based Sensor Fusion For Improved Terrain Classification." In *Autonomous Systems: Sensors, Processing, and Security for Vehicles and Infrastructure 2020* Vol. 11415, 121–128.
- Laible, S., Y. N. Khan, and A. Zell. 2013. "Terrain Classification With Conditional Random Fields On Fused 3d Lidar And Camera Data." In 2013 European Conference on Mobile Robots, 172–177.
- Lalonde, J.-F., N. Vandapel, D. F. Huber, and M. Hebert. 2006. "Natural Terrain Classification Using Three-Dimensional Ladar Data for Ground Robot Mobility." *Journal of Field Robotics* 23, no. 10: 839–861.
- Le, H.-A., T. Mensink, P. Das, S. Karaoglu, and T. Gevers. 2021. "Eden: Multimodal Synthetic Dataset of Enclosed Garden Scenes." In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, 1579–1589.
- Lee, S., H. Lim, and H. Myung. 2022. "Patchwork++: Fast and Robust Ground Segmentation Solving Partial Under-Segmentation Using 3d Point Cloud." In 2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 13276–13283.
- Lee, S.-Y., and D.-M. Kwak. 2011. "A Terrain Classification Method For UGV Autonomous Navigation Based On Surf." In 2011 8th International Conference on Ubiquitous Robots and Ambient Intelligence (URAI), 303–306. <https://doi.org/10.1109/URAI.2011.6145981>.
- Li, F., H. Zhang, H. Xu, et al. 2023. "Mask Dino: Towards a Unified Transformer-Based Framework For Object Detection and Segmentation." In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 3041–3050.
- Lim, H., M. Oh, and H. Myung. 2021. "Patchwork: Concentric Zone-Based Region-Wise Ground Segmentation With Ground Likelihood Estimation Using a 3d Lidar Sensor." *IEEE Robotics and Automation Letters* 6, no. 4: 6458–6465.
- Linnhoff, C., K. Hofrichter, L. Elster, P. Rosenberger, and H. Winner. 2022. "Measuring the Influence of Environmental Conditions on Automotive Lidar Sensors." *Sensors* 22: 5266. <https://doi.org/10.3390/s22145266>.
- Liu, X., T. Hong, Y. Yang, and Y. Gui. 2024a. "Gpnet: Point Cloud Semantic Segmentation Network For Off-Road Scenes." In 2024 China Automation Congress (CAC), 1991–1996.
- Liu, X., F. Xue, and X. Du. 2024b. "Attention Regulation For Efficient Semantic Segmentation on Unstructured Terrain." In 2024 24th International Conference on Control, Automation and Systems (ICCAS), 1529–1534.
- Liu, Y., Y. Fu, M. Qin, et al. 2024c. "Botanicgarden: A High-Quality Dataset for Robot Navigation in Unstructured Natural Environments." *IEEE Robotics and Automation Letters* 9, no. 3: 2798–2805. <https://doi.org/10.1109/LRA.2024.3359548>.
- Liu, Z., H. Tang, Y. Lin, and S. Han. 2019. "Point-Voxel CNN for Efficient 3D Deep Learning." In *Proceedings of the 33rd International Conference on Neural Information Processing Systems*, Vol. 32 (Curran Associates, Inc., 2019), pp. 965–975.
- Liyanage, D. C., R. Hudjakov, and M. Tamre. 2020. "Hyperspectral Imaging Methods Improve RGB Image Semantic Segmentation of Unstructured Terrains." In 2020 International Conference Mechatronic Systems and Materials (MSM), 1–5.
- Long, J., E. Shelhamer, and T. Darrell. 2015. "Fully Convolutional Networks For Semantic Segmentation." In *Proceedings of the IEEE Conference On Computer Vision and Pattern Recognition*, 3431–3440.
- Lu, L., C. Ordóñez, E. G. Collins, and E. M. DuPont. 2009. "Terrain Surface Classification For Autonomous Ground Vehicles Using A 2d Laser Stripe-Based Structured Light Sensor." In 2009 IEEE/RSJ International Conference on Intelligent Robots and Systems, 2174–2181. <https://doi.org/10.1109/IROS.2009.5354799>.
- Lv, F., N. Li, C. Liu, et al. 2022. "Highly Accurate Visual Method of Mars Terrain Classification for Rovers Based on Novel Image Features." *Entropy* 24, no. 9: 1304.
- Ma, X., Y. Niu, L. Gu, et al. 2021. "Understanding Adversarial Attacks on Deep Learning Based Medical Image Analysis Systems." *Pattern Recognition* 110: 107332. <https://doi.org/10.1016/j.patcog.2020.107332>.

- Maier, D., M. Bennewitz, and C. Stachniss. 2011. "Self-Supervised Obstacle Detection for Humanoid Navigation Using Monocular Vision And Sparse Laser Data." In 2011 IEEE International Conference on Robotics and Automation, 1263–1269.
- Maji, P., and R. Mullins. 2018. "On the Reduction of Computational Complexity of Deep Convolutional Neural Networks." *Entropy* 20, no. 4: 305.
- Małek, K., J. Dybała, A. Kordecki, P. Hondra, and K. Kijania. 2024. "Offroadsynth Open Dataset for Semantic Segmentation Using Synthetic-Data-Based Weight Initialization for Autonomous UGV in Off-Road Environments." *Journal of Intelligent & Robotic Systems* 110, no. 2: 1–18.
- Mars Surface Image (Curiosity Rover) Labeled Data Set—zenodo.org. 2025. <https://zenodo.org/records/1049137#.XXcCBfmsejg>.
- Martínez, J. L., J. Morales, M. Sánchez, M. Morán, A. J. Reina, and J. J. Fernández-Lozano. 2020. "Reactive Navigation on Natural Environments By Continuous Classification of Ground Traversability." *Sensors* 20, no. 22: 6423.
- Martínez-Rodríguez, J. L., and M. Morán, et al. 2020. *Supervised Learning of Natural-Terrain Traversability With Synthetic 3d Laser Scans*.
- Maturana, D., P.-W. Chou, M. Uenoyama, and S. Scherer. 2017. "Real-time Semantic Mapping for Autonomous Off-Road Navigation." In Proceedings of 11th International Conference on Field and Service Robotics (FSR '17), 335–350.
- Mayuku, O., B. W. Surgenor, and J. A. Marshall. 2021. "Multi-Resolution and Multi-Domain Analysis Of Off-Road Datasets For Autonomous Driving." In 2021 18th Conference on Robots and Vision (CRV), 165–172.
- McDaniel, M. W., T. Nishihata, C. A. Brooks, and K. Iagnemma. 2010. "Ground Plane Identification Using Lidar in Forested Environments." In 2010 IEEE International Conference on Robotics and Automation, 3831–3836.
- Meng, X., Z. Cao, S. Liang, L. Pang, S. Wang, and C. Zhou. 2018. "A Terrain Description Method for Traversability Analysis Based on Elevation Grid Map." *International Journal of Advanced Robotic Systems* 15, no. 1: 1729881417751530.
- Metzger, K., P. Mortimer, and H.-J. Wuensche. 2021. "A Fine-Grained Dataset and Its Efficient Semantic Segmentation for Unstructured Driving Scenarios." In 2020 25th International Conference on Pattern Recognition (ICPR), 7892–7899. <https://doi.org/10.1109/ICPR48806.2021.9411987>.
- Milella, A., G. Reina, J. Underwood, and B. Douillard. 2014. "Visual Ground Segmentation by Radar Supervision." *Rob Auton Syst* 62, no. 5: 696–706.
- Min, C., W. Jiang, D. Zhao, et al. 2022. "Orfd: A Dataset and Benchmark for Off-Road Freespace Detection." In 2022 International Conference On Robotics And Automation (ICRA), 2532–2538.
- Moosmann, F., O. Pink, and C. Stiller. 2009. "Segmentation of 3d Lidar Data in Non-Flat Urban Environments Using A Local Convexity Criterion." In 2009 IEEE Intelligent Vehicles Symposium, 215–220.
- Mortimer, P., R. Hagmanns, M. Granero, T. Luettel, J. Petereit, and H.-J. Wuensche. 2023. "The Goose Dataset For Perception in Unstructured Environments." arXiv preprint arXiv:2310.16788.
- Naeem, W., T. Xu, R. Sutton, and A. Tiano. 2008. "The Design of a Navigation, Guidance, and Control System for an Unmanned Surface Vehicle for Environmental Monitoring." *Proceedings of the Institution of Mechanical Engineers, Part M: Journal of Engineering for the Maritime Environment* 222, no. 2: 67–79. <https://doi.org/10.1243/14750902JEME80>.
- Nampoothiri, M. H., B. Vinayakumar, Y. Sunny, and R. Antony. 2021. "Recent Developments in Terrain Identification, Classification, Parameter Estimation for the Navigation of Autonomous Robots." *SN Applied Sciences* 3: 1–14.
- Naranjo, J. E., M. Clavijo, F. Jiménez, O. Gomez, J. L. Rivera, and M. Anguita. 2016. "Autonomous Vehicle for Surveillance Missions in Off-Road Environment." In 2016 IEEE Intelligent Vehicles Symposium (IV), 98–103.
- Neigel, P., J. Rambach, and D. Stricker. 2020. *OFFSED: Off-Road Semantic Segmentation Dataset*. <https://doi.org/10.5220/0010349805520557>.
- Neto, N. A. F., M. Ruiz, M. Reis, et al. 2022. "Low-Latency Perception in Off-Road Dynamical Low Visibility Environments." *Expert Systems With Applications* 201: 117010.
- Nie, W., P. Gou, Y. Liu, et al. 2022. "A Semi-Supervised Image Segmentation Method Based On Generative Adversarial Network." In 2022 IEEE 10th Joint International Information Technology and Artificial Intelligence Conference (ITAIC) Vol. 10: 1217–1223. <https://doi.org/10.1109/ITAIC54216.2022.9836504>.
- Noh, H., S. Hong, and B. Han. 2015. "Learning Deconvolution Network For Semantic Segmentation." In Proceedings of the IEEE International Conference on Computer Vision, 1520–1528.
- Nunes, R., J. Ferreira, and P. Peixoto. 2022. *Procedural Generation of Synthetic Forest Environments to Train Machine Learning Algorithms*.
- Omer, R., and L. Fu. 2010. "An Automatic Image Recognition System for Winter Road Surface Condition Classification." In 13th International IEEE Conference on Intelligent Transportation Systems, 1375–1379. <https://doi.org/10.1109/ITSC.2010.5625290>.
- Ono, M., T. J. Fuchs, A. Steffy, M. Maimone, and J. Yen. 2015. "Risk-Aware Planetary Rover Operation: Autonomous Terrain Classification And Path Planning." In 2015 IEEE Aerospace Conference, 1.
- Open Robotics. 2022. "Robot Operating System (ROS) 2—Humble Hawksbill." <https://ros.org/>.
- Otsu, K., M. Ono, T. J. Fuchs, I. Baldwin, and T. Kubota. 2016. "Autonomous Terrain Classification with Co- and Self-Training Approach." *IEEE Robotics and Automation Letters* 1, no. 2: 814–819. <https://doi.org/10.1109/LRA.2016.2525040>.
- Otte, S., C. Weiss, T. Scherer, and A. Zell. 2016. "Recurrent Neural Networks for Fast and Robust Vibration-Based Ground Classification on Mobile Robots." In 2016 IEEE International Conference on Robotics and Automation (ICRA), 5603–5608. <https://doi.org/10.1109/ICRA.2016.7487778>.
- Papadakis, P. 2013. "Terrain Traversability Analysis Methods for Unmanned Ground Vehicles: A Survey." *Engineering Applications of Artificial Intelligence* 26, no. 4: 1373–1385.
- Peng, J., G. Hua, S. Tian, Y. Wu, and W. Zou. 2024. "Lightweight Boundary-Assisted Network for Freespace Segmentation in Unstructured Road Scenes." *Displays* 83: 102688.
- Pfaff, P., R. Triebel, and W. Burgard. 2007. "An Efficient Extension to Elevation Maps for Outdoor Terrain Mapping and Loop Closing." *The International Journal of Robotics Research* 26, no. 2: 217–230.
- Pokonieczny, K. 2018. "Methods of Using Self-Organising Maps for Terrain Classification, Using An Example Of Developing A Military Passability Map." In Dynamics in GIScience 4, 359–371.
- Pomares, A., J. L. Martínez, A. Mandow, M. A. Martínez, M. Morín, and J. Morales. 2018. "Ground Extraction From 3d Lidar Point Clouds With the Classification Learner App." In 2018 26th Mediterranean Conference on Control and Automation (MED), 1–9. <https://doi.org/10.1109/MED.2018.8442569>.
- Pour Arab, D., M. Spisser, and C. Essert. 2024. "3D Hybrid Path Planning for Optimized Coverage of Agricultural Fields: A Novel Approach for Wheeled Robots." *Journal of Field Robotics* 42, no. 2: 455–473. <https://doi.org/10.1002/rob.22422>.
- Qingchun, F., Z. Wengang, Q. Quan, J. Kai, and G. Rui. 2012. "Study On Strawberry Robotic Harvesting System." In 2012 IEEE International Conference on Computer Science and Automation Engineering (CSAE) Vol. 1: 320–324.

- Qiu, J., and C. Jiang. 2025. "A Bilateral Semantic Guidance Network for Detection of Off-Road Freespace With Impairments Based on Joint Semantic Segmentation and Edge Detection." *Computers and Electrical Engineering* 123: 110045.
- Quinchia, A. G., G. Falco, E. Falletti, F. Dovis, and C. Ferrer. 2013. "A Comparison Between Different Error Modeling of Mems Applied to Gps/Ins Integrated Systems." *Sensors* 13, no. 8: 9549–9588.
- Rahmadian, R., and M. Widyartono. 2020. "Autonomous Robotic in Agriculture: A Review." In 2020 Third International Conference On Vocational Education And Electrical Engineering (ICVEE), 1–6.
- Rankin, A. L., A. Huertas, and L. H. Matthies. 2009. "Stereo-Vision-Based Terrain Mapping For Off-road Autonomous Navigation." In *Unmanned Systems Technology XI* Vol. 7332: 253–269.
- Reda, M., A. Onsy, A. Y. Haikal, and A. Ghanbari. 2024. "Path Planning Algorithms in the Autonomous Driving System: A Comprehensive Review." *Rob Auton Syst* 174: 104630.
- Reina, G., M. Bellone, L. Spedicato, and N. I. Giannoccaro. 2014. "3d Traversability Awareness for Rough Terrain Mobile Robots." *Sensor Review* 34, no. 2: 220–232.
- Reina, G., R. Galati, and A. Milella. 2018. "All-Terrain Estimation For Mobile Robots in Precision Agriculture." In 2018 IEEE International Conference on Industrial Technology (ICIT), 63–68.
- Reina, G., A. Milella, and R. Rouveure. 2015. "Traversability Analysis for Off-Road Vehicles Using Stereo and Radar Data." In 2015 IEEE International Conference on Industrial Technology (ICIT), 540–546. <https://doi.org/10.1109/ICIT.2015.7125155>.
- Reina, G., A. Milella, and R. Worst. 2016. "Lidar and Stereo Combination for Traversability Assessment of Off-Road Robotic Vehicles." *Robotica* 34, no.12: 2823–2841.
- Ren, K., T. Zheng, Z. Qin, and X. Liu. 2020. "Adversarial Attacks and Defenses In Deep Learning." *Engineering* 6, no. 3: 346–360. <https://doi.org/10.1016/j.eng.2019.12.012>.
- Renderdoc. 2018. <https://renderdoc.org/>.
- Rodríguez-Rodríguez, J. A., E. López-Rubio, J. A. Ángel-Ruiz, and M. A. Molina-Cabello. 2024. "The Impact of Noise and Brightness on Object Detection Methods." *Sensors* 24, no. 3: 821. <https://doi.org/10.3390/s24030821>.
- Romera, E., J. M. Alvarez, L. M. Bergasa, and R. Arroyo. 2017. "Erfnet: Efficient Residual Factorized Convnet for Real-Time Semantic Segmentation." *IEEE Transactions on Intelligent Transportation Systems* 19, no. 1: 263–272.
- Ronneberger, O., P. Fischer, and T. Brox. 2015. "U-net: Convolutional Networks for Biomedical Image Segmentation." In *Medical Image Computing and Computer-Assisted Intervention-MICCAI 2015: 18th International Conference, Munich, Germany, October 5–9, 2015, Proceedings, Part III* 18, 234–241.
- Ruff, L., R. Vandermeulen, N. Goernitz, et al. 2018. "Deep One-Class Classification." In *International Conference on Machine Learning*, 4393–4402.
- Russell, B. C., A. Torralba, K. P. Murphy, and W. T. Freeman. 2008. "Labelme: A Database and Web-Based Tool for Image Annotation." *International Journal of Computer Vision* 77: 157–173.
- Sai, C., V. Teja, A. Prudhvi, G. Kushwanth, and B. Bikash. 2024. "Terrain Classification for Enhanced Autonomous Systems." *International Journal of Research Publication and Reviews* 5: 3780–3785. <https://doi.org/10.55248/gengpi.5.0324.0776>.
- Sánchez, M., J. L. Martínez, J. Morales, A. Robles, and M. Morán. 2019. "Automatic Generation of Labeled 3d Point Clouds of Natural Environments With Gazebo." In 2019 IEEE International Conference on Mechatronics (ICM) Vol. 1: 161–166.
- Sandler, M., A. Howard, M. Zhu, A. Zhmoginov, and L.-C. Chen. 2018. "Mobilenetv2: Inverted Residuals And Linear Bottlenecks." In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 4510–4520.
- Santamaria-Navarro, À., E. H. Teniente, M. Morta, and J. Andrade-Cetto. 2015. "Terrain Classification in Complex Three-Dimensional Outdoor Environments." *Journal of Field Robotics* 32, no. 1: 42–60.
- Sathyamoorthy, A. J., K. Weerakoon, T. Guan, J. Liang, and D. Manocha. 2022. "Terrapn: Unstructured Terrain Navigation Using Online Self-Supervised Learning." In 2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 7197–7204.
- Schmid, R., D. Atha, F. Schöller, et al. 2022. "Self-Supervised Traversability Prediction by Learning to Reconstruct Safe Terrain." In 2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 12419–12425.
- Schneider, D. G., and M. R. Stemmer. 2024. "CNN-Based Multi-Object Detection and Segmentation in 3d Lidar Data for Dynamic Industrial Environments." *Robotics* 13, no. 12: 174.
- Selvathai, T., J. Varadhan, and S. Ramesh. 2017. "Road and Off Road Terrain Classification For Autonomous Ground Vehicle." In 2017 International Conference on Information Communication and Embedded Systems (ICICES), 1–3. <https://doi.org/10.1109/ICICES.2017.8070724>.
- Seo, J., T. Kim, K. Kwak, J. Min, and I. Shim. 2023a. "Scate: A Scalable Framework for Self-Supervised Traversability Estimation In Unstructured Environments." *IEEE Robotics and Automation Letters* 8, no. 2: 888–895.
- Seo, J., S. Sim, and I. Shim. 2023b. "Learning Off-Road Terrain Traversability With Self-Supervisions Only." *IEEE Robotics and Automation Letters* 8, no. 8: 4617–4624.
- Sgibnev, I., A. Sorokin, B. Vishnyakov, and Y. Vizilter. 2020. "Deep Semantic Segmentation for the Off-Road Autonomous Driving." *The International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences* 43: 617–622.
- Shaban, A., X. Meng, J. Lee, B. Boots, and D. Fox. 2022. "Semantic Terrain Classification for Off-Road Autonomous Driving." In *Conference on Robot Learning*, 619–629.
- Shan, T., J. Wang, B. Englot, and K. Doherty. 2018. "Bayesian Generalized Kernel Inference for Terrain Traversability Mapping." In *Conference on Robot Learning*, 829–838.
- Shan, Y., Y. Fu, X. Chen, et al. 2023. "Lidar Based Traversable Regions Identification Method for Off-Road UGV Driving." *IEEE Transactions on Intelligent Vehicles* 9, no. 2: 3544–3557.
- Sharma, S., J. E. Ball, B. Tang, D. W. Carruth, M. Doude, and M. A. Islam. 2019. "Semantic Segmentation With Transfer Learning for Off-Road Autonomous Driving." *Sensors* 19, no. 11: 2577.
- Sharma, S., L. Dabir, T. Hannis, et al. 2022. "Cat: Cava Traversability Dataset for Off-Road Autonomous Driving." *IEEE Access* 10: 24759–24768. <https://doi.org/10.1109/ACCESS.2022.3154419>.
- Shi, X., S. Shao, T. Wang, and Q. Zhang. 2023. "A Terrain Recognition Method Based on Semantic Segmentation for Field Robot Under Sample Imbalance." In 2023 IEEE International Conference on Unmanned Systems (ICUS), 1053–1058.
- Singh, A., K. Singh, and P. Sujit. 2021. "Offroadtranseg: Semi-Supervised Segmentation Using Transformers on Offroad Environments." arXiv preprint arXiv:2106.13963.
- Singh, P., R. Chukkappalli, S. Chaudhari, et al. 2024. "Shifting to Machine Supervision: Annotation-Efficient Semi and Self-Supervised Learning for Automatic Medical Image Segmentation and Classification." *Scientific Reports* 14, no. 1: 10820.
- Singh, U., K. Saurabh, N. Trehan, R. Vyas, and O. Vyas. 2022. *Terrain Classification Using Transfer Learning on Hyperspectral Images: A Comparative Study*.

- Society of Automotive Engineers. 2021. "Taxonomy and Definitions for Terms Related to Driving Automation Systems for On-Road Motor Vehicles (sae standard j3016_202104)." https://www.sae.org/standards/content/j3016_202104/.
- Statistics and Machine Learning Toolbox—mathworks.com. 2025. <https://www.mathworks.com/products/statistics.html>.
- Suger, B., B. Steder, and W. Burgard. 2015. "Traversability Analysis for Mobile Robots in Outdoor Environments: A Semi-Supervised Learning Approach Based on 3d-Lidar Data." In 2015 IEEE International Conference on Robotics and Automation (ICRA), 3941–3946. <https://doi.org/10.1109/ICRA.2015.7139749>.
- Sumathi, T., C. L. Devasena, and M. Hemalatha. 2011. "An Overview of Automated Image Annotation Approaches." *International Journal of Research and Reviews in Information Sciences* 1, no. 1: 1–5.
- Sung, G.-Y., D.-M. Kwak, and J. Lyou. 2010. "Neural Network Based Terrain Classification Using Wavelet Features." *Journal of Intelligent & Robotic Systems* 59: 269–281.
- Swan, R. M., D. Atha, H. A. Leopold, et al. 2021. "Ai4mars: A Dataset for Terrain-Aware Autonomous Driving on Mars." In 2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW), 1982–1991. <https://doi.org/10.1109/CVPRW53098.2021.00226>.
- Synthetic Data Generation Market to Skyrocket to 2.1 Billion by 2028—bccresearch.com. 2025. <https://www.bccresearch.com/pressroom/ift/synthetic-data-generation-market-to-skyrocket-to-21-billion-by-2028>.
- Szabó, L., and Z. Weltsch. 2024. "A Comprehensive Review of Existing Datasets for Off-Road Autonomous Vehicles." In 2024 IEEE 22nd World Symposium on Applied Machine Intelligence and Informatics (SAMI), 000403–000410.
- The Most Powerful Real-Time 3D Creation Tool—unrealengine.com. 2025. <https://www.unrealengine.com/en-US>.
- The Perception Camera Component, Perception Package, 1.0.0-preview.1—1.0. 2025. <https://docs.unity3d.com/Packages/com.unity.perception@1.0/manual/PerceptionCamera.html>.
- Tobin, J., R. Fong, A. Ray, J. Schneider, W. Zaremba, and P. Abbeel. 2017. "Domain Randomization for Transferring Deep Neural Networks From Simulation to the Real World." In 2017 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 23–30.
- Tola, E., V. Lepetit, and P. Fua. 2010. "Daisy: An Efficient Dense Descriptor Applied to Wide-Baseline Stereo." *IEEE Transactions on Pattern Analysis and Machine Intelligence* 32, no. 5: 815–830. <https://doi.org/10.1109/TPAMI.2009.77>.
- Torii, T. 2000. "Research in Autonomous Agriculture Vehicles in Japan." *Computers and Electronics in Agriculture* 25, no. 1: 133–153. [https://doi.org/10.1016/S0168-1699\(99\)00060-5](https://doi.org/10.1016/S0168-1699(99)00060-5).
- Triebel, R., P. Pfaff, and W. Burgard. 2006. "Multi-Level Surface Maps for Outdoor Terrain Mapping and Loop Closing." In 2006 IEEE/RSJ International Conference on Intelligent Robots and Systems, 2276–2282.
- Truong, J., S. Chernova, and D. Batra. 2021. "Bi-Directional Domain Adaptation for sim2real Transfer of Embodied Navigation Agents." *IEEE Robotics and Automation Letters* 6, no. 2: 2634–2641.
- Unity Real-Time Development Platform, 3D, 2D, VR & AR Engine—unity.com. 2025. <https://unity.com/>.
- Upcroft, B., M. Ridley, L. L. Ong, et al. 2008. "Multi-Level State Estimation in an Outdoor Decentralised Sensor Network." In Experimental Robotics: The 10th International Symposium on Experimental Robotics, 355–365.
- Valada, A., R. Mohan, and W. Burgard. 2020. "Self-Supervised Model Adaptation for Multimodal Semantic Segmentation." *International Journal of Computer Vision* 128, no. 5: 1239–1285.
- Valada, A., G. L. Oliveira, T. Brox, and W. Burgard. 2017. "Deep Multispectral Semantic Scene Understanding of Forested Environments Using Multimodal Fusion." In 2016 International Symposium On Experimental Robotics, 465–477.
- Varma, M., and A. Zisserman. 2002. "Classifying Images of Materials: Achieving Viewpoint and Illumination Independence." In Computer Vision—ECCV 2002: 7th European Conference on Computer Vision Copenhagen, Denmark, May 28–31, 2002 Proceedings, Part III 7, 255–271.
- Vaswani, A., N. Shazeer, N. Parmar, et al. 2017. "Attention Is All You Need (Version 7)." Preprint, arXiv, June 12. <https://doi.org/10.48550/ARXIV.1706.03762>.
- Vecherin, S. N., J. R. Desmond, T. S. Hodgdon, et al. 2020. Artificial Intelligence and Machine Learning for Autonomous Military Vehicles.
- Vernaza, P., B. Taskar, and D. D. Lee. 2008. "Online, Self-Supervised Terrain Classification via Discriminatively Trained Submodular Markov Random Fields." In 2008 IEEE International Conference on Robotics and Automation, 2750–2757.
- Vishnyakov, B., Y. Blokhinov, I. Sgibnev, et al. 2020. "Semantic Scene Understanding for the Autonomous Platform." *International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences* 43: 637–644.
- Viswanath, K., K. Singh, P. Jiang, P. Sujit, and S. Saripalli. 2021. "Offseg: A Semantic Segmentation Framework for Off-Road Driving." In 2021 IEEE 17th International Conference on Automation Science and Engineering (CASE), 354–359.
- Vosviewer - Visualizing Scientific Landscapes. 2010. <https://www.vosviewer.com/>.
- Vu, H., H. T. Nguyen, P. M. Chu, et al. 2017. "Adaptive Ground Segmentation Method for Real-Time Mobile Robot Control." *International Journal of Advanced Robotic Systems* 14, no. 6: 1729881417748135.
- Walch, W., M. Eder, K. Mautner-Lassnig, and G. Steinbauer-Wagner. 2022. "Offroad Terrain Classification for Mobile Robots." In Proceedings of the ARW 2022, 6–11.
- Wang, H., X. Jiang, H. Ren, Y. Hu, and S. Bai. 2021. "Swiftnet: Real-Time Video Object Segmentation." In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 1296–1305.
- Wang, H., E. Lu, X. Zhao, and J. Xue. 2023. "Vibration and Image Texture Data Fusion-Based Terrain Classification Using Wknn for Tracked Robots." *World Electric Vehicle Journal* 14, no. 8: 214.
- Wang, J., K. Sun, T. Cheng, et al. 2020. "Deep High-Resolution Representation Learning for Visual Recognition." *IEEE Transactions on Pattern Analysis and Machine Intelligence* 43, no. 10: 3349–3364.
- Wang, P. 2021. "Research on Comparison of Lidar and Camera in Autonomous Driving." In Journal of Physics: Conference Series Vol. 2093, 012032.
- Wellhausen, L., A. Dosovitskiy, R. Ranftl, K. Walas, C. Cadena, and M. Hutter. 2019. "Where Should I Walk? Predicting Terrain Properties From Images via Self-Supervised Learning." *IEEE Robotics and Automation Letters* 4, no. 2: 1509–1516.
- Wellhausen, L., R. Ranftl, and M. Hutter. 2020. "Safe Robot Navigation via Multi-Modal Anomaly Detection." *IEEE Robotics and Automation Letters* 5, no. 2: 1326–1333.
- Wigness, M., S. Eum, J. G. Rogers, D. Han, and H. Kwon. 2019. "A Rugs Dataset for Autonomous Navigation and Visual Perception in Unstructured Outdoor Environments." In 2019 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 5000–5007.
- Wijayathunga, L., A. Rassau, and D. Chai. 2023. "Challenges and Solutions for Autonomous Ground Robot Scene Understanding and Navigation in Unstructured Outdoor Environments: A Review." *Applied Sciences* 13, no. 17: 9877.
- Winkens, C., V. Kobelt, and D. Paulus. 2017. "Robust Features for Snapshot Hyperspectral Terrain-Classification." In Computer Analysis

- of Images and Patterns: 17th International Conference, CAIP 2017, Ystad, Sweden, August 22–24, 2017, Proceedings, Part I 17, 16–27.
- Wolf, D. F., G. S. Sukhatme, D. Fox, and W. Burgard. 2005. “Autonomous Terrain Mapping and Classification Using Hidden Markov Models.” In Proceedings of the 2005 IEEE International Conference on Robotics and Automation, 2026–2031.
- Wu, B., A. Wan, X. Yue, and K. Keutzer. 2018. “Squeezeseg: Convolutional Neural Nets With Recurrent CRF For Real-Time Road-Object Segmentation From 3d Lidar Point Cloud.” In 2018 IEEE International Conference On Robotics and Automation (ICRA), 1887–1893.
- Wu, L., and T. Liu. 2024. “Drivable Area Detection in Off-Road for Autonomous Driving.” In 2024 IEEE 7th International Conference on Electronic Information and Communication Technology (ICEICT), 515–520.
- Yan, T., X. Zheng, W. Liu, B. Liang, and Z. Chen. 2022. “The Synthetic Off-Road Trail Dataset for Unmanned Motorcycle.” In 2022 IEEE 95th Vehicular Technology Conference:(VTC2022-Spring), 1–7.
- Yang, Z., Y. Tan, S. Sen, et al. 2022. “Uncertainty-Aware Perception Models for Off-Road Autonomous Unmanned Ground Vehicles.” arXiv preprint arXiv:2209.11115.
- Yeong, D. J., G. Velasco-Hernandez, J. Barry, and J. Walsh. 2021. “Sensor and Sensor Fusion Technology in Autonomous Vehicles: A Review.” *Sensors* 21, no. 6: 2140. <https://doi.org/10.3390/s21062140>.
- Yu, C., C. Gao, J. Wang, G. Yu, C. Shen, and N. Sang. 2021. “Bisenet v2: Bilateral Network With Guided Aggregation for Real-Time Semantic Segmentation.” *International Journal of Computer Vision* 129: 3051–3068.
- Yu, Z. 2020. “Research on Remote Sensing Image Terrain Classification Algorithm Based On Improved Knn.” In 2020 IEEE 3rd International Conference on Information Systems and Computer Aided Education (ICISCAE), 569–573. <https://doi.org/10.1109/ICISCAE51034.2020.9236884>.
- Yuksel, Y., N. El-Sheimy, and A. Nouredin. 2010. “Error Modeling and Characterization of Environmental Effects for Low Cost Inertial MEMS Units.” In IEEE/ION Position, Location and Navigation Symposium, 598–612. <https://doi.org/10.1109/PLANS.2010.5507180>.
- Zhang, H., X. Dai, F. Sun, and J. Yuan. 2016. “Terrain Classification in Field Environment Based on Random Forest for The Mobile Robot.” In 2016 35th Chinese Control Conference (CCC), 6074–6079.
- Zhang, H., H. Ma, Q. Mao, X. Xusheng, and C. Wang. 2024. “Key Technology of Temporary Support Robot for Rapid Excavation of Coal Mine Roadway.” *Journal of Field Robotics* 42. <https://doi.org/10.1002/rob.22416>.
- Zhang, K. Z. 2024. “Applications and Prospects of AI in Autonomous Cars—Take Tesla as an Example.” In 2nd International Conference on Mechatronic Automation and Electrical Engineering (ICMAEE 2024) Vol. 2024: 355–360. <https://doi.org/10.1049/icp.2024.4527>.
- Zhang, L., A. Zhu, S. Zhao, and Y. Zhou. 2021. “Simulation of Atmospheric Visibility Impairment.” *IEEE Transactions on Image Processing* 30: 8713–8726. <https://doi.org/10.1109/tip.2021.3120044>.
- Zhao, H., J. Shi, X. Qi, X. Wang, and J. Jia. 2017. “Pyramid Scene Parsing Network.” In Proceedings of the IEEE Conference On Computer Vision and Pattern Recognition, 2881–2890.
- Zhou, S., J. Xi, M. W. McDaniel, T. Nishihata, P. Saleses, and K. Iagnemma. 2012. “Self-Supervised Learning to Visually Detect Terrain Surfaces for Autonomous Robots Operating in Forested Terrain.” *Journal of Field Robotics* 29, no. 2: 277–297.
- Zhou, X., Y. Feng, X. Li, Z. Zhu, and Y. Hu. 2023. “Off-Road Environment Semantic Segmentation for Autonomous Vehicles Based on Multi-Scale Feature Fusion.” *World Electric Vehicle Journal* 14, no. 10: 291.
- Zou, W., G. Hua, G. Chen, et al. 2023. “Need a Dog for Seeing Eye? A Walk Viewpoint Dataset for Freespace Detection in Unstructured Environments.” In 2023 IEEE International Conference on Multimedia and Expo (ICME), 1541–1546.
- Zou, Y., W. Chen, L. Xie, and X. Wu. 2014. “Comparison of Different Approaches to Visual Terrain Classification for Outdoor Mobile Robots.” *Pattern Recognition Letters* 38: 54–62. <https://doi.org/10.1016/j.patrec.2013.11.004>.
- Zürn, J., W. Burgard, and A. Valada. 2020. “Self-Supervised Visual Terrain Classification From Unsupervised Acoustic Feature Learning.” *IEEE Transactions on Robotics* 37, no. 2: 466–481.